

CHAPTER

1

What Is a Project?

Things are not always what they seem.

— Phaedrus, Roman writer and fabulist

CHAPTER LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- ◆ Define a project
- ◆ List a project's characteristics
- ◆ Distinguish a project from a program, activity, and task
- ◆ Understand the five parameters that constrain a project
- ◆ Know the importance of defining and using a project classification rule
- ◆ Understand the issues around scope creep, hope creep, effort creep, and feature creep

To put projects into perspective, you need a definition — a common starting point. All too often, people call any work they have to do a “project.” Projects actually have a very specific definition. If a set of tasks or work to be done does not meet the strict definition, then it cannot be called a project. To use the project management techniques presented in this book, you must first have a project.

Defining a Project

A *project* is a sequence of unique, complex, and connected activities that have one goal or purpose and that must be completed by a specific time, within budget, and according to specification.

This definition tells you quite a bit about a project. To appreciate just what constitutes a project, take a look at each part of the definition.

Sequence of Activities

A project comprises a number of activities that must be completed in some specified order, or *sequence*. An *activity* is a defined chunk of work.

CROSS-REF Chapter 4 expands on this informal definition of an activity.

The sequence of the activities is based on technical requirements, not on management prerogatives. To determine the sequence, it is helpful to think in terms of the following inputs and outputs:

- What is needed as input in order to begin working on this activity?
- What activities produce those deliverables as output?

The output of one activity or set of activities becomes the input to another activity or set of activities.

Specifying a sequence based on resource constraints or statements such as “Pete will work on activity B as soon as he finishes working on activity A” should be avoided because this establishes an artificial relationship between activities. What if Pete wasn’t available at all? Resource constraints aren’t ignored when you actually schedule activities. The decision of what resources to use and when to use them comes later in the project planning process.

Unique Activities

The activities in a project must be *unique*. A project has never happened exactly in the same way before, and it will never happen again under the same conditions. Something is always different each time the activities of a project are repeated. Usually the variations are random in nature — for example, a part is delayed, someone is sick, or a power failure occurs. These are random events that can happen, but you never are sure of when or how, and what impact they will have on the schedule. These random variations are the challenge for the project manager.

Complex Activities

The activities that make up the project are not simple, repetitive acts, such as mowing the lawn, painting the house, washing the car, or loading the delivery truck. They are *complex*. For example, designing an intuitive user interface to an application system is a complex activity.

Connected Activities

Connectedness implies that there is a logical or technical relationship between pairs of activities. There is an order to the sequence in which the activities that make up the project must be completed. They are considered connected because the output from one activity is the input to another. For example, you must design the computer program before you can program it.

You could have a list of unconnected activities that must all be complete in order to complete the project. For example, consider painting the interior rooms of a house. With some exceptions, the rooms can be painted in any order. The interior of a house is not completely painted until all its rooms have been painted, but they may be painted in any order. Painting the house is a collection of activities, but it is not considered a project according to the definition.

One Goal

Projects must have a single *goal* — for example, to design an inner-city playground for ADC (Aid to Dependent Children) families. However, very large or complex projects may be divided into several *subprojects*, each of which is a project in its own right. This division makes for better management control. For example, subprojects can be defined at the department, division, or geographic level. This artificial decomposition of a complex project into subprojects often simplifies the scheduling of resources and reduces the need for interdepartmental communications while a specific activity is worked on. The downside is that the projects are now interdependent. Even though interdependency adds another layer of complexity and communication, it can be handled.

Specified Time

Projects have a specified *completion date*. This date can be self-imposed by management or externally specified by a client or government agency. The deadline is beyond the control of anyone working on the project. The project is over on the specified completion date whether or not the project work has been completed.

Within Budget

Projects also have *resource limits*, such as a limited amount of people, money, or machines that are dedicated to the project. These resources can be adjusted up or down by management, but they are considered *fixed resources* by the project manager. For example, suppose a company has only one web designer at the moment. That is the fixed resource that is available to project managers. Senior management can change the number of resources, but that luxury is not available to the project manager. If the one web designer is fully scheduled, the project manager has a resource conflict that he or she cannot resolve.

CROSS-REF Chapter 6 covers resource limits in more detail.

According to Specification

The client, or the recipient of the project's deliverables, expects a certain level of functionality and quality from the project. These expectations can be self-imposed, such as the specification of the project completion date, or client-specified, such as producing the sales report on a weekly basis.

Although the project manager treats the specification as fixed, the reality of the situation is that any number of factors can cause the specification to change. For example, the client may not have defined the requirements completely, or the business situation may have changed (which often happens in projects with long durations). It is unrealistic to expect the specification to remain fixed through the life of the project. Systems specification can and will change, thereby presenting special challenges to the project manager.

CROSS-REF Chapters 8, 11, and 12 describe how to effectively handle client requirements.

What Is a Program?

A *program* is a collection of related projects. The projects must be completed in a specific order for the program to be considered complete. Because programs comprise multiple projects, they are larger in scope than a single project. For example, the United States government has a space program that includes several projects such as the Challenger project. A construction company contracts a program to build an industrial technology park with several separate projects.

Unlike projects, programs can have many goals. For example, every launch of a new mission in the NASA space program includes several dozen projects in the form of scientific experiments. Except for the fact that they are all aboard

the same spacecraft, the experiments are independent of one another and together define a program.

Establishing Temporary Program Offices

As the size of the project increases, it becomes unwieldy from a management standpoint. A common practice is to establish a temporary program office to manage these large projects. One of my clients uses a team size of 30 as the cutoff point. Whenever the team size is greater than 30, a program office is established. That program office consists of nothing more than the management structure needed for the project. There will be a program director and one or more program administrators as support. The program administrators support the program manager as well as the teams. Even for teams of 30, there will often be a subteam organization put in place to simplify the management of the team. Each subteam will be led by a subproject manager. When the project is completed, the program office disbands.

Establishing Permanent Program Offices

A permanent program office is established to manage an ongoing and changing portfolio of projects. The portfolio consists of projects that have something in common — for example, all might be funded from the same budget, might be linked to the same goal statement, or might use the same resource pool. The permanent program office, unlike the temporary program office, manages a continuously changing collection of projects.

Understanding the Scope Triangle

The following five constraints operate on every project:

- Scope
- Quality
- Cost
- Time
- Resources

These constraints form an interdependent set — a change in one constraint can require a change in another constraint in order to restore the equilibrium of the project. In this context, the set of five parameters form a system that must remain in balance for the project to be in balance. Because they are so important to the success or failure of the project, each parameter is discussed individually in this section.

Scope

Scope is a statement that defines the boundaries of the project. It tells not only what will be done but also what will not be done. In the information systems industry, scope is often referred to as a *functional specification*. In the engineering profession, it is generally called a *statement of work*. Scope may also be referred to as a document of understanding, a scoping statement, a project initiation document, or a project request form. Whatever its name, this document is the foundation for all project work to follow. It is critical that the scope be correct. Chapter 3 describes exactly how this should happen in its coverage of Conditions of Satisfaction (COS).

Beginning a project on the right foot is important, and so is staying on the right foot. It is no secret that a project's scope can change. You do not know how or when, but it will change. Detecting that change and deciding how to accommodate it in the project plan are major challenges for the project manager.

CROSS-REF Chapter 3 is devoted to defining project scope, and scope management is discussed in Chapter 6.

Quality

The following two types of quality are part of every project:

- **Product quality** — The quality of the deliverable from the project. The traditional tools of quality control, discussed in Chapter 2, are used to ensure product quality.
- **Process quality** — The quality of the project management process itself. The focus is on how well the project management process works and how can it be improved. Continuous quality improvement and process quality management are the tools used to measure process quality. These are discussed in Chapter 15.

A sound quality management program with processes in place that monitor the work in a project is a good investment. Not only does it contribute to client satisfaction, it helps organizations use their resources more effectively and efficiently by reducing waste and revisions. Quality management is one area that should not be compromised. The payoff is a higher probability of successfully completing the project and satisfying the client.

Cost

The dollar cost of doing the project is another variable that defines the project. It is best thought of as the budget that has been established for the project.

This is particularly important for projects that create deliverables that are sold either commercially or to an external customer.

Cost is a major consideration throughout the project management life cycle. The first consideration occurs at an early and informal stage in the life of a project. The client can simply offer a figure about equal to what he or she had in mind for the project. Depending on how much thought the client put into it, the number could be fairly close to or wide of the actual cost for the project. Consultants often encounter situations in which the client is willing to spend only a certain amount for the work. In these situations, you do what you can with what you have. In more formal situations, the project manager prepares a proposal for the projected work. That proposal includes an estimate (perhaps even a quote) of the total cost of the project. Even if a preliminary figure has been supplied by the project manager, the proposal allows the client to base his or her go/no-go decision on better estimates.

Time

The client specifies a time frame or deadline date within which the project must be completed. To a certain extent, cost and time are inversely related to one another. The time a project takes to be completed can be reduced, but costs increase as a result.

Time is an interesting resource. It can't be inventoried. It is consumed whether you use it or not. The objective for the project manager is to use the future time allotted to the project in the most effective and productive ways possible. Future time (time that has not yet occurred) can be a resource to be traded within a project or across projects. Once a project has begun, the prime resource available to the project manager to keep the project on schedule or get it back on schedule is time. A good project manager realizes this and protects the future time resource jealously.

CROSS-REF Chapters 4 and 5, which discuss scheduling project activities, cover this topic in more detail.

Resources

Resources are assets such as people, equipment, physical facilities, or inventory that have limited availabilities, can be scheduled, or can be leased from an outside party. Some are fixed; others are variable only in the long term. In any case, they are central to the scheduling of project activities and the orderly completion of the project.

For systems development projects, people are the major resource. Another valuable resource for systems projects is the availability of computer processing

time (mostly for testing purposes), which can present significant problems to the project manager with regard to project scheduling.

Envisioning the Scope Triangle as a System in Balance

Projects are dynamic systems that must be kept in equilibrium. Not an easy task, as you shall see! Figure 1-1 illustrates the dynamics of the situation.

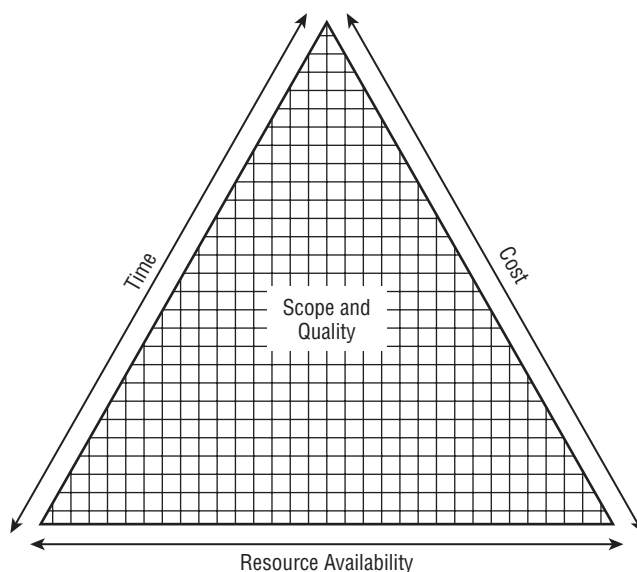


Figure 1-1 The scope triangle

The geographic area inside the triangle represents the scope and quality of the project. Lines representing time, cost, and resource availability bound scope and quality. Time is the window of time within which the project must be completed. Cost is the dollar budget available to complete the project. Resources are any consumables used on the project. People, equipment availability, and facilities are examples.

NOTE While the accountants will tell you that everything can be reduced to dollars, and they are right, you will separate resources as defined here. They are controllable by the project manager and need to be separately identified for that reason.

The project plan will have identified the time, cost, and resource availability needed to deliver the scope and quality of a project. In other words, the project is in equilibrium at the completion of the project planning session and approval of the commitment of resources and dollars to the project. That will not last too long, however. Change is waiting around the corner.

The scope triangle offers a number of insights into the changes that can occur in the life of the project. For example, the triangle represents a system in balance before any project work has been done. The sides are long enough to encompass the area generated by the scope and quality statements. Not long after work begins, something is sure to change. Perhaps the client calls with an additional requirement for a feature that was not envisioned during the planning sessions. Perhaps the market opportunities have changed, and it is necessary to reschedule the deliverables to an earlier date, or a key team member leaves the company and is difficult to replace. Any one of these changes throws the system out of balance.

The project manager controls resource utilization and work schedules. Management controls cost and resource level. The client controls scope, quality, and delivery dates. Scope, quality, and delivery dates suggest a hierarchy for the project manager as solutions to accommodate the changes are sought.

CROSS-REF Chapters 5 and 6 discuss this topic in greater detail.

Managing the Creeps

While some of your team members may occasionally seem like creeps to you, that is not creep management I am talking about. *Creeps* here refer to minute changes in the project due to the obscure, and for awhile unnoticeable, actions of team members. Many of these go undetected until a problem raises its ugly head. There are four types of creeps, as described in the sections that follow.

Scope Creep

Scope creep is the term that has come to mean any change in the project that was not in the original plan. Change is constant. To expect otherwise is simply unrealistic. Changes occur for several reasons that have nothing to do with the ability or foresight of the client, the project manager, or a project team member. Market conditions are dynamic. The competition can introduce or announce an upcoming new version of its product. Your management might decide that getting the product to market before the competition is necessary. Scope creep isn't necessarily anyone's fault. It is just a reality that has to be

CHAPTER

4

How to Plan a Project

This report, by its very length, defends itself against the risk of being read.

– Winston Churchill, English Prime Minister

The man who goes alone can start today, but he who travels with another must wait 'til that other is ready.

– Henry David Thoreau, American naturalist

Every moment spent planning saves three or four in execution.

– Crawford Greenwalt, President, DuPont

The hammer must be swung in cadence, when more than one is hammering the iron.

– Giordano Bruno, Italian philosopher

CHAPTER LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- ◆ Understand the importance of planning a project
- ◆ Understand the purpose of the Joint Project Planning Session (JPPS)
- ◆ Know how to plan a JPPS
- ◆ Know the contents of the project proposal
- ◆ Recognize the difference between activities and tasks
- ◆ Understand the importance of the completeness criteria to your ability to manage the work of the project

(continued)

CHAPTER LEARNING OBJECTIVES (Continued)

- ◆ Explain the approaches to building the Work Breakdown Structure (WBS)
- ◆ Generate a complete WBS
- ◆ Use the WBS as a planning tool and reporting tool
- ◆ Understand top-down versus bottom-up processes for building the WBS
- ◆ Define a work package and its purposes
- ◆ Understand the difference between effort and duration
- ◆ Explain the relationship between resource loading and task duration
- ◆ List and explain the causes of variation in task duration
- ◆ Use any one of six task-duration estimation methods
- ◆ Understand the process of creating cost estimates at the task level
- ◆ Schedule people to project activities using a skill matrix
- ◆ Understand the process of determining resource requirements at the task level
- ◆ Construct a network representation of the project tasks
- ◆ Understand the four types of task dependencies and when they are used
- ◆ Recognize the types of constraints that create task sequences
- ◆ Compute the earliest start (ES), earliest finish (EF), latest start (LS), and latest finish (LF) times for every task in the network
- ◆ Understand lag variables and their uses
- ◆ Identify the critical path in the project
- ◆ Define free slack and total slack and know their significance
- ◆ Analyze the network for possible schedule compression
- ◆ Use advanced network dependency relationships for improving the project schedule
- ◆ Understand and apply management reserve
- ◆ Utilize various approaches to leveling resources
- ◆ Determine the appropriate use of substitute resources

How often have you heard it said that planning is a waste of time? No sooner is the plan completed than someone comes along to change it. These same naysayers would also argue that the plan, once completed, is disregarded and merely put on the shelf so the team can get down to doing some real work. As this chapter points out, these views are incorrect.

Tools, Templates, and Processes Used to Plan a Project

If you were able to do a project twice — once with a good plan and once with a poor or no plan — the project with the good plan would finish earlier, including the time spent planning. The project with a good plan has a higher probability of finishing than does the poorly planned project. The quality is better, the cost is less, and the list of benefits to good planning goes on. So why is planning often seen as not being real work? Figure 4-1 expresses my message clearer than mere words could.

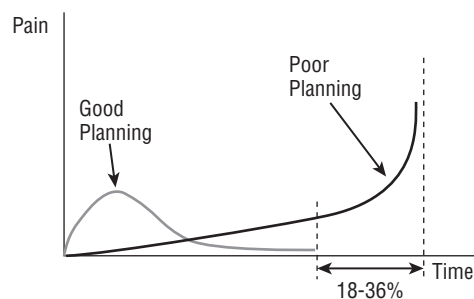


Figure 4-1 Pain curves

“Pay me now or pay me later” applies equally well to the oil change commercial as it does to project planning. When the team and management are anxious for work to begin, it is difficult to focus on developing a solid plan of action before you are pressed into service. At times it would seem that the level of detail in the plan is overkill, but it is not. The project manager must resist the pressure to start project work and instead spend the time up front generating a detailed project plan. It has been demonstrated that a poor planning effort takes its toll later in the project as schedules slip, quality suffers, and expectations are not met.

The pain curve demonstrates that proper planning is painful but pays off in less pain later in the project. To not plan is to expose yourself to significant pain as the project proceeds. In fact, that pain usually continues to increase. It would continue to increase indefinitely except that someone usually pulls the plug on the project when the pain reaches unbearable levels.

In this chapter, you will learn all of the tools, templates, and processes that you will need to generate good project plans. More specifically, you will learn about the following:

- Planning and running a Joint Project Planning Session (JPPS)
- Creating the Work Breakdown Structure (WBS)

- Techniques for estimating task duration, resource requirements, and cost
- Building the initial schedule
- Analyzing the initial schedule
- Schedule compression techniques
- Risk management planning
- Writing an effective project proposal
- Gaining approval of the project proposal

The Importance of Planning

If you are to be an effective project manager, a project plan is indispensable. Not only is it a road map to how the work is scheduled, but it is also a tool to aid in your decision making. The plan suggests alternative approaches, schedules, and resource requirements from which you can select the best alternative.

NOTE Understand that a project plan is dynamic. It is a statement of intent, not a statement of fact. You *expect* it to change. A complete plan will clearly state the tasks that need to be done, why they are necessary, who will do what, when the project will be completed, what resources will be needed, and what criteria must be met in order for the project to be declared complete and successful. However, traditional project management (TPM) models are not designed for change, even though it is expected. Part II of this book describes project management life cycle (PMLC) models that are designed for change. One of the many advantages of these models is that change is accommodated within the process itself. Change in the TPM world is something the project manager would rather not deal with, whereas the project manager who is using the models discussed in Part II sees change as a necessary ingredient of a successful project.

There are three benefits to developing a project plan. They are

1. Planning Reduces Uncertainty

Even though you would never expect the project work to occur exactly as planned, planning the work enables you to consider the likely outcomes and to put the necessary corrective measures in place when things don't happen according to plan.

2. Planning Increases Understanding

The mere act of planning gives you a better understanding of the goals and objectives of the project. Even if you were to discard the plan, you would still benefit from having done the exercise.

3. Planning Improves Efficiency

After you have defined the project plan and the necessary resources to carry out the plan, you can schedule the work to take advantage of resource availability. You also can schedule work in parallel — that is, you can do tasks concurrently, rather than in series. By doing tasks concurrently, you can shorten the total duration of the project. You can maximize your use of resources and complete the project work in less time than by taking other approaches.

Just as Alice needed to know where in Wonderland she was going, the project manager needs to know the goal to be achieved by the project and the steps that will be taken to attain that goal, i.e., the solution. Not knowing the parameters of a project prevents measurement of progress and results in never knowing when the project is complete. The plan also provides a basis for measuring work planned against work performed.

Using Application Software Packages to Plan a Project

Before I begin discussing the tools, templates, and processes needed for project planning, I want spend some time on the software packages and why you might or might not use them. I have always been an advocate of using the appropriate tools to plan a project. My experiences have ranged from the back of the napkin to the use of sophisticated modeling and prototyping tools. The size of the project has a lot to do with your choice of software packages. The larger the project, the more you will need to depend on software packages. But what about small projects or projects that are done in some incremental or iterative fashion? The answer is not always clear, but the following section describes my approach.

Determining the Need for a Software Package?

Project management software packages (at least those priced under \$1000 per seat) are both a boon and a bust to project teams. On the boon side of the ledger, they are great planning tools and allow the project manager to investigate several alternatives without the accompanying labor of having to manually adjust the planning parameters. On the bust side, they have not been very helpful in managing resources and in fact have had some rather bizarre results.

Schedule updates are also a troublesome area. The problem lies in getting reliable estimates of percent complete and estimated time to completion from each task manager. Garbage in, garbage out. These data are essential to

maintaining the project plan. On balance, the project manager has to be aware of the time savings, the time drains, and the reliability of the schedule updates in deciding whether to use a package or not. It all comes down to the value that is added for the effort that is expended. The smaller the project, the less likely it is that you will find added value in using the software package. Clearly for large or even medium-sized projects, software packages are a must.

For the types of projects that this book deals with, the answers aren't all that obvious. In these cases, project software packages can make the task of building the initial project plan a bit less labor-intensive than manual alternatives are. But even that is a matter of choice and preference. You can just as easily move sticky notes across a whiteboard as you can drag and drop task nodes across a PERT chart. For schedule updating, I have a clear preference for the sticky notes. My reasoning is that I can get the entire team involved in the exercise, and everyone can see the alternatives much easier on the whiteboard than they can on the computer screen. For distributed teams, some compensation can be made by using web-based tools to communicate sticky note information electronically.

The best testimony I have is from my own experiences managing a three-year, \$5M project. All I used were the planning tools discussed in the next section. The agile project was completed nine months early, successfully and significantly under budget.

Project Planning Tools

The list here is very short: sticky notes, marking pens, and plenty of whiteboard space. I don't want you to think that I have taken a step backward to a time when there were no automated tools. Quite the contrary: I do use automated tools for project planning and execution. The tools I just mentioned happen to be my choice for some incremental, most iterative, all adaptive, and all extreme projects. These are introduced in Chapter 8 and discussed in detail in Chapters 9, 11, and 12. That accounts for over 80 percent of all projects. The reason is simple. These projects all proceed in short cycles of two to four weeks. You don't need a sledge hammer to kill a mosquito! If you really have to depend on software tools, go ahead. Just remember that if you create the plan using software tools, you have to maintain it.

Sticky Notes

Sticky notes are used to record information about a single task in the project. The information that you might want to record on a sticky note includes the following:

- Task ID
- Unique task name

- Task duration
- Task labor
- Resource requirements
- Task manager
- Calculated values such as earliest start (ES), earliest finish (EF), latest start (LS), and latest finish (LF)
- Critical path (calculated)

Color-coded sticky notes offer a number of alternatives for the creative planner. For example, you can use a different color to represent each of the following:

- The type of task (critical, for example)
- Specific parts of the WBS (design, build, test, and implement, for example)
- A position on the team (a critical or scarce skill, for example)

Using sticky notes in this way is not only visually appealing, but it's very informative during resource scheduling and finalization of the project plan. With experience, the color coding becomes intuitive.

Marking Pens

For the purposes of project planning, you will need dry-erase marking pens. They come in several colors, but you will only need the black and red ones. They are used to visually display the dependencies (black marking pens) that exist among and between the project tasks or the critical path (red marking pens).

Whiteboard

The whiteboard is indispensable. Flip charts are not a good alternative. For large projects, you need to have a minimum of about 30 linear feet of whiteboard space for planning purposes. The room that provides this space will become the team war room and should be reserved for the exclusive use of the team for the entire project. It will need to be a secure room as well.

CROSS-REF See Chapter 5 for more on the team war room.

The whiteboard will be used to create, document, and in some cases post the following:

- Project Overview Statement (POS)*
- WBS*

- Dependency diagram*
- Initial project schedule
- Final project schedule*
- Resource schedule*
- Issues log*
- Updated project schedule*

The items shown with an asterisk (*) are permanently posted on the whiteboard and updated as required.

Portable electronic whiteboards can be used when a dedicated space is not available.

How Much Time Should Planning Take?

This is one of those “it all depends” questions. This is not a question that is easily answered because there are a number of variables that will affect planning time. The most important variables are project complexity, solution clarity, and the availability of the team members and the client for planning meetings. The actual labor involved in building a plan for the typical small project is about one workday. Ideally that would occur in one calendar day, but peoples’ schedules might make that impossible. When you have difficulty scheduling the planning team, don’t revert to planning by walking around. That just doesn’t work. Trust me — I learned the hard way!

As a rule of thumb the following estimates can be applied:

Very small projects: Less than 1/2 day

Small projects: Less than one day

Medium projects: 2 days

Large projects: 3–4 days

Very large projects: 30 team members translates to a large project. The planning time can vary widely from 5 or more days to several months.

Running the Planning Session

Consider the ideal situation first. All of the following activities are part of the planning process, and all of them are completed in one workday. This section gives you a high-level look at what these activities are. In subsequent sections, you’ll learn the details of how they are to be accomplished.

In a one-day planning session for a typical small project, the planning team performs the following major activities:

- Reviews the POS for clarity
- Creates the complete WBS, including the Activity List
- Estimates task duration and resource needs
- Constructs project network diagram
- Determines critical path
- Revises and approves project completion date
- Finalizes resource schedule
- Gains consensus on the project plan

The project manager can run the planning session for the small project. For larger or more complex projects, it pays to have someone other than the project manager facilitate the planning meetings. To complete this planning session in one day, the project manager will have to tightly control the discussion and keep the planning team moving forward. Any briefing materials that can be distributed ahead of time will help reduce briefing time in the actual planning meeting. There should be a timed agenda, and everyone must commit to sticking to it. Ground rules need to be put in place. One time-saving alternative is to have the project manager and the client complete the Conditions of Satisfaction (COS) and Project Overview Statement (POS) ahead of time and circulate these documents to the planning team prior to the meeting.

Planning and Conducting Joint Project Planning Sessions

All of the planning activities discussed so far to create the detailed project plan take place in a Joint Project Planning Session (JPPS). I advocate and use a group process for generating the detailed project plan. The JPPS is a group session in which all of the people who are involved in the project meet to develop the detailed plan. The session can last from one to three days, and it can be work-intensive. Conflict between session attendees is common, but the final result of this meeting is an agreement about how the project can be accomplished within a specified time frame, budget, resource availabilities, and according to client requirements.

NOTE My planning process shares many of the same features as Joint Requirements Planning (JRP) and Joint Applications Design (JAD) sessions. The JRP session is commonly used to design computer applications. My JPPS is robust — that is, it can be used for any type of project, including the design and development of computer applications.

The objective of a JPPS is this: Develop a project plan that meets the COS as negotiated between the requestor and the provider, and as described in the POS. Sounds simple, doesn't it?

Unfortunately, that agreement doesn't often happen with any regularity, for many reasons. The client and the project team are generally impatient to get on with the work of the project. After all, there are deadlines to meet and other projects demanding the team member's attention. Team members don't have time for planning — there is too much work to do and too many clients to satisfy. Regrettably, at the project's eleventh hour, when it is too late to recover from a poor plan, the team and the client bow in defeat. Next time, pay more attention to the planning details. But somehow that next time never seems to come. It's time for change!

In this day and age, the virtual team seems to be the rule rather than the exception. To accommodate this type of team, the project manager usually does one-on-one planning with each team member and consolidates the results for review with the entire team participating in an online review session.

Planning the JPPS

Team planning has always been viewed as advantageous over other forms of project planning, such as the project manager planning the project by walking around gathering data for the plan. In my experience, the synergy of the group provides far more accurate activity duration estimates and more complete information input to the planning process itself. Perhaps the best advantage of all is that it creates a much stronger commitment to the project on the part of all those who lived through the pain of generating and agreeing to the complete project plan. If all else fails, it is more fun than doing planning in isolation.

I know you sometimes feel that planning is a necessary evil. It is something you do because you have to and because you can then say that you have thought about where you want to go and how you are going to get there. After they have been written, plans are often bound in nice notebooks and become bookends gathering dust on someone's shelf or in a file folder in your desk drawer. Make up your mind right now to change that!

Planning is essential to good project management. The plan that you generate is a dynamic document. It changes as the project commences. It will be a reference work for you and the team members when questions of scope and change arise. Make no bones about it: To do good planning is painful,

but to do poor planning is even more painful. Remember the pain curves in Figure 4-1? Which one will you choose?

The first document considered in the JPPS is the POS. One may already exist and therefore will be the starting point for the JPPS. If one doesn't exist, it must be developed as the initial part of or as a prerequisite to starting the JPPS. The situation will dictate how best to proceed. The POS can be developed in a number of ways. If it is an idea for consideration, it will probably be developed by one individual — typically the person who will be the project manager. It can be departmentally based or cross-departmentally based. The broader the impact on the enterprise, the more likely it will be developed as the first phase of a JPPS. Finally, the POS may have been developed through a COS exercise. In any case, the JPPS begins by discussing and clarifying exactly what is intended by the POS. The project team might also use this opportunity to write the Project Definition Statement (PDS) — their understanding of the project. The PDS is nothing more than an expanded version of the POS, but from the perspective of the planning team.

The JPPS must be planned down to the last detail if it is to be successful. Time is a scarce resource for all of us, and the last thing you want to do is waste it. Recognize before you start that the JPPS will be very intense. Participants often get emotional and will even dig their heels in to make a point.

Before learning about how to plan and conduct a JPPS, take a look at who should attend.

Attendees

The JPPS participants are invited from among those who might be affected by or have input into the project. If the project involves deliverables or is a new process or procedure, then anyone who has input to the process, receives output from the process, or handles the deliverables should be invited to participate in the JPPS. The client falls into one or more of these categories and must be present at the JPPS. Any manager of resources that may be required by the project team should also attend the JPPS. In many organizations, the project has a project champion (not necessarily the project manager or client manager) who may wish to participate at least at the start.

Here is a list of potential JPPS attendees:

Facilitator — A successful JPPS requires an experienced facilitator. This person is responsible for conducting the JPPS. It is important that the facilitator not have a vested interest or bring biases to the session because that would diminish the effectiveness of the plan. It must be developed with an open mind, not with a biased mind. For this reason, I strongly suggest that the project manager not facilitate the session. If using an outside consultant is not possible, I recommend that you select a neutral party to act as the facilitator, such as another project manager.

Project manager — Because you are not leading the planning session, you can concentrate on the plan itself, which is your major role in the JPPS. Even if you received the assignment before any planning has been done, having you facilitate the JPPS may seem to be an excellent option, but it can be the wrong choice if the project is politically charged or has clients from more than one function, process, or resource pool. You must be comfortable with the project plan. After all, you are the one who has final responsibility when it comes to getting the project done on time, within budget, and according to specification.

Another project manager — Skilled JPPS facilitators are hard to find. Since you are not a good choice for facilitator, then maybe another project manager — presumably unbiased — would be a good choice, especially if he or she has JPPS experience.

JPPS consultant — Project management consultants will often serve as another source of qualified JPPS facilitators. Their broad experience in project management and project management consulting will be invaluable. This is especially true in organizations that have recently completed project management training and are in the process of implementing their own project management methodology. Having an outside consultant facilitate the JPPS is as much a learning experience as it is an opportunity to get off to a good start with a successful JPPS.

Technographer — The JPPS facilitator is supported by a technographer, a professional who not only knows project management but is also an expert in the software tools used to document the project plan. While the JPPS facilitator is coordinating the planning activities, the JPPS technographer is recording planning decisions on the computer as they occur in real time. At any point in time — and there will be several — the technographer can print out or display the plan for all to see and critique.

Core project team — Commitment is so important that to exclude the core team from the JPPS would be foolish. Estimating activity duration and resource requirements will be much easier with the professional expertise these people can bring to the planning session. The core project team is made up of individuals (both from the client and from the provider) who will stay with the project from the first day to the last day. This does not mean that they are with the project full-time. In today's typical organization, an individual would not be assigned to only one project at a time, unless the organization is totally projectized or uses self-directed teams.

Client representative — This attendee is always a bit tricky. Face it: Some clients really don't want to be bothered. It is up to the project manager or

champion to convince clients of the importance of their participation in the JPPS. I don't claim that this will be easy, but it is nevertheless important. The client must buy in to the project plan. The client won't have that buy-in if the project manager simply mails a copy of the plan. The client must be involved in the planning session. To proceed without the client's involvement is to court disaster. Changes to the project plan will occur, and problems will arise. If the client is involved in preparing the plan, he or she can contribute to resolutions of change requests and problem situations.

Resource managers — These managers control resources that the project will require. Putting a schedule together without input and participation from these managers would be a waste of time. They may have some suggestions that will make the plan more realistic, too. In some cases, they may send a representative who will also be part of the project team. The important factor here is that someone from each resource area is empowered to commit resources to the project plan. These are not commitments to provide a specific named person or room. They are commitments to provide a certain skill set or type of facility.

Project champion — The project champion drives the project and sells it to senior management. In many cases, the champion can be the client — which is an ideal situation because the client is already committed to the project. In other cases, the project champion can be the senior managers of the division, department, or process that will be the beneficiary of the project deliverables.

Functional managers — Because functional managers manage areas that can either provide input to or receive output from the project deliverables, they or a representative should participate in the planning session. They will ensure that the project deliverables can be smoothly integrated into existing functions or that the functions will have to be modified as part of the project plan.

Process owner — For the same reasons that functional managers should be present, so should process owners. If the project deliverables do not smoothly integrate into their processes, either the project plan or the affected processes will have to be altered.

A formal invitation that announces the project, its general direction and purpose, and the planning schedule should be issued by the project manager to all of the other attendees.

NOTE RSVPs are a must! Full attendance is so important that I have canceled the JPPS when certain key participants were not able to attend. On one occasion, I acted as the project manager for a client and cancelled the JPPS because the client did not think his attendance was important enough. My feedback to the client was that as soon as it was a high enough priority for him to attend, I would reschedule the JPPS. Pushback like this is tough, but the JPPS is so critically important to the ultimate success of the project that I was willing to take this strong position with the client.

Facilities

Because the planning team may spend as many as three consecutive days in planning, it is important that the physical facility is comfortable and away from the daily interruptions. To minimize distractions, you might be tempted to have the planning session offsite. However, I prefer onsite planning sessions. Onsite planning sessions have both advantages and disadvantages, but with proper preparation, they can be controlled. In my experience, having easy access to information is a major advantage to onsite planning sessions, but interruptions due to the daily flow of work are a major disadvantage. With easy access to the office made possible by cell phones and e-mail, the potential for distraction and interruptions has increased. These need to be minimized in whatever way makes sense.

Allocate enough space so that each group of four or five planning members can have a separate work area with a table, chairs, and a flip chart. All work should be done in one room. In my experience, breakout rooms tend to be dysfunctional. To the extent possible, everybody needs to be present for everything that takes place in the planning session. The room should have plenty of whiteboard space or blank walls. In many cases, I have taped flip-chart paper or butcher paper to the walls. You can never have enough writing space in the planning room.

Equipment

You will need an ample supply of sticky notes, tape, scissors, and colored marking pens. For more high-tech equipment, an LCD projector and a PC are all you need for everyone in the room to see the details as they come together.

The Complete Planning Agenda

The agenda for the JPPS is straightforward. It can be completed in one, two, or three sessions. For example, an early meeting with the requestor can be scheduled, at which time the COS are drafted. These will be input to the

second session, during which the POS is drafted. In cases where the POS must be approved before detailed planning can commence, there will be an interruption until approval can be granted. After approval is obtained, the third session can be scheduled. At this session (which is usually two or three days long), the detailed project plan can be drafted for approval.

Here's a sample agenda for the JPPS:

Session 1

1. Negotiate the COS.
2. Gather requirements.

Session 2

1. Write the POS.

Session 3 (JPPS)

1. The entire planning team creates the first-level WBS.
2. Subject matter experts develop further decomposition, with the entire planning team observing and commenting.
3. Estimate activity durations and resource requirements.
4. Construct a project network diagram.
5. Determine the critical path.
6. Revise and approve the project completion date.
7. Finalize the resource schedule.
8. Gain consensus on the project plan.

Deliverables

The deliverables from the JPPS are provided in the PMLC. They are listed here:

Work Breakdown Structure Recall that the WBS is a graphical or indented outline list of the work (expressed as activities) to be done to complete the project. It is used as a planning tool as well as a reporting structure.

Activity Duration Estimates The schedule, which is also a major deliverable, is developed from estimates of the duration of each work activity in the project. Activity duration estimates may be single-point estimates or three-point estimates, as discussed later in this chapter.

Resource Requirements For each activity in the project, an estimate of the resources to perform the work is required. In most cases, the resources will be the technical and people skills, although they can also include such things as physical facilities, equipment, and computer cycles.

Project Network Schedule Using the WBS, the planning team will define the sequence in which the project activities should be performed. Initially, this sequence is determined only by the technical relationships between activities, not by management prerogatives. That is, the deliverables from one or more activities are needed to begin work on the next activity. You can understand this sequence most easily by displaying it graphically. The definition of the network activities and the details of the graphical representation are covered later in this chapter.

Activity Schedule With the sequence determined, the planning team will schedule the start date and end date for each activity. The availability of resources will largely determine that schedule.

Resource Assignments The output of the activity schedule will be the assignment of specific resources (such as skill sets) to the project activities.

Project Notebook Documentation can be a chore to produce. But that's not the case in the five-phase PMLC described in this book, where project documentation is a natural by-product of the project work. All you need to do is appoint a project team member to be responsible for gathering information that is already available, putting it in a standard format, and electronically archiving it. This responsibility begins with the project planning session and ends when the project is formally closed.

Conducting the JPPS

The first priority of the facilitator is to create an open and collaborative environment for the planning team. There is going to be disagreement, and all members of the planning team must feel free to express their thoughts. In conducting the sessions, the facilitator must encourage everyone to fully participate. Those who are more reserved must be drawn into the conversation by the facilitator. Likewise, those who tend to dominate the conversation must be diplomatically controlled by the facilitator. Excellent meeting management skills are required. That is why a trained facilitator is preferred over a project manager when it comes to running a JPPS.

Building the Work Breakdown Structure

The WBS is a hierarchical description of all work that must be done to complete the project as defined in the POS. Several processes can be used to create this hierarchy, as described in this section.

The Requirements Breakdown Structure (RBS) is the input to the WBS construction process. If the RBS is complete, then a traditional approach to project management can be taken. This chapter describes how to build a complete WBS. However, in most cases the WBS will not be complete, and some other project management approach will have to be taken. Those are discussed later in Chapters 11 and 12.

An example of the WBS is shown in Figure 4-2.

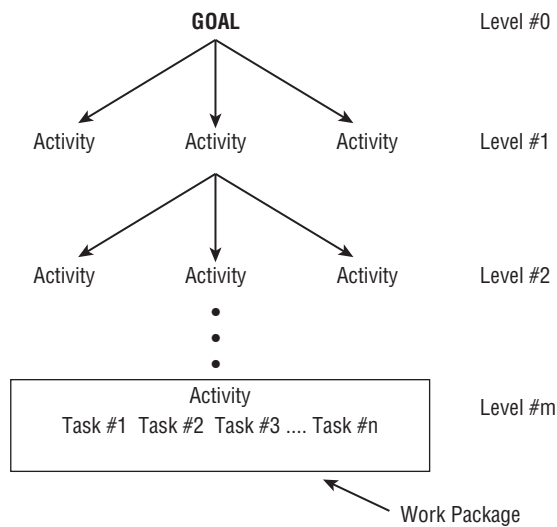


Figure 4-2 Hierarchical visualization of the WBS

To understand the WBS, you need to be familiar with the terms introduced in Figure 4-2. The first term is *activity*. An activity is simply a chunk of work. (Later in this chapter, the “Six Criteria to Test for Completeness in the WBS” section expands on this definition.) The second term is *task*. Note that in Figure 4-2, activities turn to tasks at some level in the hierarchy. A task is a smaller chunk of work. Although these definitions may seem a bit informal, the difference between an activity and a task will become clearer shortly.

The terms “activity” and “task” have been used interchangeably among project managers and project management software packages. Some think of activities as being made up of tasks, others say that tasks are made up of activities, and still others use one term to represent both concepts. In this book, I refer to higher-level work as activities. An activity is composed of two or more tasks. When the tasks that make up an activity are complete, the activity is complete.

Another term is *work package*. A work package is a complete description of how the tasks that make up an activity will actually be done. It includes a

description of the what, who, when, and how of the work. Work packages are described in more detail in Chapter 5.

Breaking down work into a hierarchy of activities, tasks, and work packages is called *decomposition*. For example, take a look at the top of the WBS in Figure 4-2. Notice that the goal statement from the POS is defined as a *Level 0* activity in the WBS. The next level, Level 1, is a decomposition of the Level 0 activity into a set of activities defined as *Level 1* activities. These Level 1 activities are major chunks of work. When the work associated with each Level 1 activity is complete, the Level 0 activity is complete. For this example, that means the project is complete. As a general rule, when an activity at Level n is decomposed into a set of activities at Level $n + 1$ and the work associated with those activities is complete, the activity at Level n , from which they were defined, is complete.

Decomposition is important to the overall project plan because it enables you to estimate the duration of the project, determine the required resources, and schedule the work. The complete decomposition is developed by using the completeness criteria discussed later in this chapter. By following those criteria, activities at the lowest levels of decomposition will possess known properties that enable you to meet planning and scheduling needs.

This process of decomposition is analogous to the process many students use in school for writing research papers. Despite the teacher's extolling the value of preparing a detailed outline before writing the paper, the student chooses to do it the other way around — writing the paper first and extracting the outline from it. That won't work in project planning. You have to define the work before you set out to do it.

Those who have experience in systems development should see the similarity between hierarchical decomposition and functional decomposition. In principle, there is no difference between a WBS and a functional decomposition of a system. My approach to generating a WBS departs from the generation of a functional decomposition in that I follow a specific process with a stopping rule for completing the WBS. I am not aware of a similar process for generating the functional decomposition of a system. Veterans of systems development might even see some similarity to older techniques such as stepwise refinement or pseudo-code. These tools do, in fact, have a great deal in common with the techniques I use to generate the WBS.

Uses for the WBS

The WBS has four uses: as a thought-process tool, an architectural-design tool, a planning tool, and a project-status-reporting tool. The following sections describe how to use the WBS for each of these purposes.

Thought-Process Tool

First, and maybe foremost, the WBS is a thought process. As a thought process, it is a design and planning tool. It helps the project manager and the planning team visualize exactly how the work of the project can be defined and managed effectively. It would not be unusual to consider alternative ways of decomposing the work until an alternative is found with which the project manager is comfortable.

Architectural-Design Tool

When all is said and done, the WBS is a picture of the work of the project and how the items of work are related to one another. It must make sense. In that context, it is a design tool.

Planning Tool

In the planning phase, the WBS gives the planning team a detailed representation of the project as a collection of activities that must be completed in order for the project to be completed. It is at the lowest activity level of the WBS that you will estimate effort, elapsed time, and resource requirements; build a schedule of when the work will be completed; and estimate deliverable dates and project completion.

Project-Status-Reporting Tool

Although this is not a common use of the WBS, it has been used as a structure for reporting project status. The project activities are consolidated (that is, rolled up) from the bottom as lower-level activities are completed. As work is completed, activities will be completed. Completion of lower-level activities causes higher-level activities to be partially complete. Shading is often used to highlight completed tasks and activities. Some of these higher-level activities may represent significant progress whose completion will be milestone events in the course of the project. Thus, the WBS defines milestone events that can be reported to senior management and the client.

NOTE Trying to find a happy compromise between a WBS architecture that lends itself well to the planning thought process and the rolling up of information for summary reporting can be difficult. It is best to have input from all the parties that may be using the WBS before settling on a design. There is no one right way to do it; it's subjective. You will get better with practice.

In the final analysis, it is the project manager who decides on the architecture of the WBS and the level of detail required. This detail is important because the project manager is accountable for the success of the project. The WBS must be defined so that the project manager can manage the project. That means that the approach and detail in the WBS might not be the way others would have approached it. Apart from any senior management requirements for reporting or organizational requirements for documentation or process, the project manager is free to develop the WBS according to his or her needs and those of management. Because of this requirement, the WBS is not unique. That should not bother you because all that is required is a WBS that defines the project work so that you, the project manager, can manage it. “Beauty is in the eyes of the beholder” applies equally well to the choice of which of several approaches to building the WBS is the best choice. As project manager you have the responsibility of making that choice.

Generating the WBS

The best way to generate the WBS is as part of the JPPS. You’ll learn the steps as you look at two different approaches to building the WBS in this section. Before I discuss those approaches, I want to remind you of where you are in the planning process and then offer a few general comments about procedures I have followed in my practice regardless of the approach taken.

One of the two simple decomposition processes is used to identify the activities that must be performed from the beginning to the completion of the project. These activities are the lowest level of managed work for the project manager. At this point in the planning process, you should have completed the POS. You may have to go back and reconsider the POS as a result of further planning activities, but for now assume the POS is complete. My technique for generating the WBS will reduce even the most complex project to a set of clearly defined activities. The WBS will be the document that guides the remainder of the planning activities.

As many as 10 to 20 participants may be involved in building the WBS, so gathering around a computer screen won’t do the job. Neither will projecting the screen on an overhead LCD projector. The only way I have found that works consistently is to use sticky-notes, marking pens, and plenty of whiteboard space. In the absence of whiteboard space, you might wallpaper the planning room with flip-chart or butcher paper. You cannot have too much writing space. Using butcher paper, I have even filled the four walls of the planning room and several feet of hallway outside the planning room. It is sloppy, but it gets the job done.

Two approaches can be used to identify the project activities. The first is the top-down approach; the second is the bottom-up approach.

Top-Down Approach

The top-down approach begins at the goal level (which may be a deliverable) and successively partitions work down to lower levels of decomposition until the participants are satisfied that the work has been sufficiently defined. The completion criteria discussed later in this chapter structure the partitioning exercise for this approach.

After the project activities have been defined using the top-down approach and they satisfy the completion criteria given later in this chapter, they are defined at a sufficient level of detail to enable you to estimate time, cost, and resource requirements first at the activity level and then aggregate to the project level. Because the activities are defined to this level of detail, the project time, cost, and resource requirements are estimated much more accurately.

Once the activities are described, you can sequence the project work so that as many activities as possible are performed in parallel, rather than in sequence. In other words, the list of activities can be sequenced so that the project duration (clock time needed to complete all project work) will be much shorter than the sum of all the times needed to complete each activity.

I recommend two variations of the top-down approach: the team approach and the subteam approach. I have used both in my consulting practice.

Team Approach

Although it requires more time to complete than the subteam approach (discussed next), the team approach is the better of the two. In this approach, the entire team works on all parts of the WBS. For each Level 1 activity, appoint the most knowledgeable member of the planning team to facilitate the further decomposition of that part of the WBS. Continue with similar appointments until the WBS is complete. This approach enables all members of the planning team to pay particular attention to the WBS as it is developed, noting discrepancies in real time.

Subteam Approach

When time is at a premium, the planning facilitator will prefer the subteam approach. The first step is to divide the planning team into as many subteams as there are activities at Level 1 of the WBS. Then follow these steps:

1. The planning team agrees on the approach to building the first level of the WBS.
2. The planning team creates the Level 1 activities.
3. A subject matter expert leads the team in further decomposition of the WBS for his or her area of expertise.
4. The team suggests decomposition ideas for the expert until each activity within the Level 1 activities meets the WBS completion criteria.

Note that the entire planning team decides on the approach for the first-level breakdown. After that, the group is partitioned into subteams, with each subteam having some expertise for that part of the WBS. It is hoped that they will have all the expertise they need to develop their part of the WBS. If not, outside help may be brought in as needed. Be careful not to clutter the team with too many people.

It is important to pay close attention to each presentation and ask yourself these questions: Is there something in the WBS that I did not expect to see? Is there something missing from the WBS that I expected to see? The focus here is to strive for a complete WBS. In cases where the WBS will be used for reporting purposes, the project manager must be careful to attach lower-level activities to higher-level activities to preserve the integrity of the status reports that will be generated.

As the discussion continues and activities are added and deleted from the WBS, questions about agreement between the WBS and the POS will occur. Throughout the exercise, the POS should be posted on flip-chart paper and hung on the walls of the planning room. Each participant should compare the scope of the project as described in the POS with the scope as presented in the WBS. If something in the WBS appears out of scope, challenge it. Either redefine the scope or discard the appropriate WBS activities. Similarly, look for complete coverage of the scope as described in the WBS with the POS. This is the time to be critical and carefully define the scope and work to accomplish it. Mistakes found now, before any work is done, are far less costly and disruptive than they will be if found late in the project.

The dynamic at work here is one of changing project boundaries. Despite all efforts to the contrary, the boundaries of the project are never clearly defined at the outset. There will always be reasons to question what is in and what is not in the project. That is fine. Just remember that the project boundaries have not yet been set in concrete. That will happen after the project has been approved to begin. Until then, you are still in the planning mode.

Bottom-Up Approach

Another method for identifying the activities in the project is to take a bottom-up approach. This approach is more like a brainstorming session than an organized approach to building the WBS.

The first steps in the bottom-up approach are the same as those for the top-down approach. Namely, the entire planning team agrees to the first-level breakdown. The planning team is then divided into as many groups as there are first-level activities. Each group makes a list of the activities that must

be accomplished in order to complete the first-level activity. To do this, they proceed as follows:

1. Someone in the group identifies an activity and announces it to the group. If the group agrees, then the activity is written on a slip of paper and put in the middle of the table. The process repeats itself until no new ideas are forthcoming.
2. The group sorts the slips into activities that seem to be related to one another. This grouping activity should help the planning team add missing activities or remove redundant ones.
3. When the team is satisfied it has completed the activity list for the first-level breakdown, the members are finished. Each group then reports to the entire planning team the results of its work.
4. Final critiques are given, missing activities are added, and redundant activities are removed.

WARNING Although I have seen this approach work in a few cases, there is the danger of not defining all activities or of defining activities at too high or low a level of granularity. The completeness criteria that I define later in the chapter are not ensured through this process. As a result, you may not have as manageable a project as you would if you followed the top-down approach. Obviously, risk is associated with the bottom-up approach — if you do not have to take the risk, why expose yourself to it voluntarily? The bottom-up approach randomly identifies activities and tasks, which increases the possibility of missing something. On the other hand, the top-down approach is a logically organized approach, and the likelihood of missing an activity or task is minimal. Unless there is a compelling reason to the contrary, I recommend the top-down approach. In my experience there is less danger of missing part of the project work when you use the top-down approach.

Using the WBS for Large Projects

For very large projects, you may be tempted to modify the top-down approach. Although I prefer to avoid modification, difficulty in scheduling people for the planning meeting may necessitate some modification. This section describes a modification to the top-down approach that serves this situation well.

As project size increases, it becomes unwieldy to build the entire WBS with the all of the planning team assembled. When the size of the project forces you into this situation, begin by decomposing the WBS down to Level 3. At that

point, develop intermediate estimates of time, resources, and dependencies for all Level 3 activities. The planning session is adjourned, and the Level 3 activity managers are charged with completing the WBS for their part of the project. They will convene a JPPS to complete that work. The JPPS facilitator may choose to consolidate the separate Level 3 WBS documents into a WBS for the entire project. Then the full JPPS team can be reassembled, and the planning process continues from that point.

In many large projects, the project manager may manage only down to the Level 3 activities, leaving each Level 3 activity manager to manage his or her part of the project according to the schedule developed at Level 3.

Iterative Development of the WBS

After the RBS has been built, you choose the best-fit PMLC. Depending on your feeling about the degree of completeness of the RBS, you will choose one of the five approaches: Linear, Incremental, Iterative, Adaptive, Extreme, or Emertxe. As discussed earlier in the chapter, the less you know about the RBS and the less certain you are that the client has been able to define the solution, the more likely you are to choose an approach toward the Adaptive and Extreme part of the landscape. The Linear and Incremental approaches require a complete WBS. The Iterative approach can work in cases where all of the functionality has been defined and only a few features have not. These features will be described at the lower levels of decomposition of the WBS, but they will not be completely defined. When there are missing or incomplete functions, an Adaptive approach should be used, and the WBS will have large gaps that need to be defined through iteration. Finally, when even the goal is not clearly defined, very little of the WBS can be built at the beginning of the project. Both the goal and solution (functions and features) must be defined through iteration.

Six Criteria to Test for Completeness in the WBS

Developing the WBS is the most critical part of the JPPS. If you do this part right, the rest is comparatively easy. How do you know that you've done this right? Each activity must possess the following six characteristics in order for the WBS to be deemed to be complete — that is, completely decomposed. When an activity has reached that status, it changes from an activity to a task. The six characteristics that an activity must possess to be called a task are as follows:

- Status and completion are measurable.
- The activity is bounded.
- The activity has a deliverable.

- Time and cost are easily estimated.
- Activity duration is within acceptable limits.
- Work assignments are independent.

If the activity does not possess all six of these characteristics, decompose the activity and check it again at that next lower level of decomposition. As soon as an activity possesses the six characteristics, there is no need to further decompose it. As soon as every activity in the WBS possesses these six characteristics, the WBS is defined as complete. The following sections look at each of these characteristics in more detail.

NOTE In 1991, Joseph Weiss and I introduced a four-criteria version of this completion test in *5-Phase Project Management: A Practical Planning and Implementation Guide* (Perseus Publishing Company, 1992). I have since continued to refine the criteria in the completion test to include the six characteristics presented here.

Status and Completion Are Measurable

The project manager can question the status of an activity at any point in time during the project. If the activity has been defined properly, that question is answered easily. For example, if a system's documentation is estimated to be about 300 pages long and requires approximately four months of full-time work to write, here are some possible reports that your activity manager could provide regarding the status:

- The activity is supposed to take four months of full-time work. I've been working on it for two months full-time. I guess I must be 50 percent complete.
- I've written 150 pages, so I guess I am 50 percent complete.
- I've written and had approved 150 pages and estimate that the remaining work will require two more months. I am 50 percent complete.

No one would buy the first answer, but how many times is that the information a project manager gets? Even worse, how many times does the project manager accept it as a valid statement of progress? Although the second answer is a little better, it doesn't say anything about the quality of the 150 pages that have been written, nor does it say anything about the re-estimate of the remaining work. You can see that an acceptable answer must state what has been actually completed (that is *approved*, not just written) and what remains to be done, along with an estimate to completion. Remember that you'll always know more tomorrow than you do today. After working through about half of the activity, the activity manager should be

able to give a very accurate estimate of the time required to complete the remaining work.

A simple metric that has met with some success is to compute the proportion of tasks completed as a percentage of all tasks that make up the activity. For example, if the activity has six tasks associated with it and four of the tasks are complete, the ratio of tasks completed to total tasks is 4/6 — that is, the activity is 60 percent complete. Even if work is done on the fifth task in this activity, because the task is not complete on the report date, it cannot be counted in the ratio. This metric certainly represents a very objective measure. Although it isn't completely accurate, it is a good technique. Best of all, it's quick. A project manager and activity manager do not have to sit around mired in detail about the percentage complete. You can use this same approach to measure the earned value of an activity.

CROSS-REF Earned value is defined and discussed in Chapter 6.

The Activity Is Bounded

Each activity should have a clearly defined start and end event. After the start event has occurred, work can begin on the activity. The deliverable is most likely associated with the end event that signals work is closed on the activity. For example, the start event for systems documentation might be notifying the team member who will manage the documentation creation that the final acceptance tests of the system are complete. The end event would be notifying the project manager that the client has approved the systems documentation.

The Activity Has a Deliverable

The result of completing the work that makes up the activity is the production of a deliverable. The deliverable is a visible sign that the activity is complete. This sign could be an approving manager's signature, a physical product or document, the authorization to proceed to the next activity, or some other sign of completion. The deliverable from an activity is output from that activity, which then becomes input to one or more other activities that follow its completion.

Time and Cost Are Easily Estimated

Each activity should have an estimated time and cost of completion. Being able to do this at the lowest level of decomposition in the WBS enables you to aggregate to higher levels and estimate the total project cost and the completion date. By successively decomposing activities to finer levels of granularity, you are likely to encounter primitive activities that you have performed before. This experience at lower levels of definition gives you a stronger base on which to estimate activity cost and duration for similar activities.

Activity Duration Is Within Acceptable Limits

Although there is no fixed rule for the duration of an activity, I recommend that an activity have a duration of less than two calendar weeks. This seems to be a common practice in many organizations. Even for long projects in which contractors may be responsible for major pieces of work, they will generate plans that decompose their work to activities with a two-week or shorter duration. Exceptions will occur when the activity defines process work, as is the case in many manufacturing situations. In addition, there will be exceptions for activities involving work that's repetitive and simple. For example, if you are going to build 500 widgets and it takes 10 weeks to complete this activity, you need not decompose it into five activities with each one building 100 widgets. This type of activity requires no further breakdown. If you can estimate the time to check one document, then it does not make much difference if the activity requires two months to check 400 documents or four two-week periods to check 100 documents per period. The danger you avoid is longer-duration activities whose delay can create a serious project-scheduling problem.

Work Assignments Are Independent

It is important that each activity be independent. An activity should continue reasonably well without interruption and without the need for additional input or information until the activity is complete. The work effort could be contiguous, but it can be scheduled otherwise for a variety of reasons. You can choose to schedule it in parts because of resource availability, but you could have scheduled it as one continuous stream of work.

Related to activity independence is the temptation to micromanage an activity. Best practices suggest that you manage an individual's work down to units of one week. For example, Harry is going to work on an activity that will require 10 hours of effort. The activity is scheduled to begin on Monday morning and be completed by Friday afternoon. Harry has agreed that he can accommodate the 10 hours within the week, given his other commitments that same week. Harry's manager (or the project manager) could ask Harry to report exactly when during the week he will be working on this 10-hour activity and then hold him to that plan, but what a waste of everyone's time that would be! Why not give Harry credit for enough intelligence to manage his commitments at the one-week level? No need to drill down into the workweek and burden Harry with a micro-plan and his manager with the task of managing that micro-plan. Such a scenario may in fact increase the time to complete the activity because it has been burdened with unnecessary management overhead.

The Seventh Criteria for Judging Completeness

I've separated this from the preceding six criteria because it is not a criterion in the same sense as they are. This seventh "criteria" is pure judgment on the part of the project manager. A WBS could be defined as complete based on the preceding six criteria, yet the project manager might have a lingering doubt simply because of the way the client conducted themselves during the WBS decomposition process. Something might alert the project manager that things may not be what they seem to be. For example, perhaps the client never seemed to be fully engaged or never bought into the decomposition process. They simply went along with the exercise, but never really contributed to it. That might give you reason to suspect that scope change requests may be just around the corner and that you had better have chosen a project management approach that accommodates change, rather than one that incorrectly assumes the WBS to be complete. Better to be safe than sorry. If there is any doubt in your mind about the completeness of the WBS, choose some type of Iterative or Adaptive approach so that you will be in a position to accept scope change requests.

Exceptions to the Completion Criteria Rule

In some cases, the completion criteria do not have to be satisfied in order for the WBS to be considered complete. Two common scenarios are discussed in the sections that follow.

Stopping Before Completion Criteria Are Met

A common situation where this occurs is duration-related. Suppose the activity calls for building 100 widgets, and it takes one day per widget. The maximum duration for a task has been set at 10 days. If you follow that rule, you would decompose the activity to build 100 widgets into 10 activities. That doesn't add any value to the WBS — it only adds management time. Leave the activity at 100 days and simply ask for status at the appropriate intervals. Adding activities that simply increase management time and do not add value to the project are a waste of time.

Decomposing Beyond Completion of the Criteria

For projects of a shorter duration (for example, four weeks), it would be poor management to set the acceptable activity duration at 10 days. That would create a situation with too few management checkpoints and hence create a project that would be poorly managed. Instead, the acceptable duration limit might be set at three days, for example. Even shorter duration limits might be imposed for projects that contain surgical procedures or other processes of very short duration. If the entire surgical procedure lasts only a few hours,

the acceptable duration limit might be set at a few minutes. These will always be judgment calls on your part. Do what your common sense tells you to do rather than conform to a rule that might not be appropriate given the situation.

One other situation often arises when decomposition beyond satisfaction of the completion criteria makes sense. That is with activities that are considered high risk or have a high estimated duration variance. An activity with an eight-day duration and five-day variance should raise your concern. Even though the activity has met the duration limit of 10 days, you should further decompose it in an attempt to isolate high-risk parts or the high variance parts of the activity. Again, do what makes sense.

Approaches to Building the WBS

There are many ways to build the WBS. Even though you might like the choice to be a personal one that you, the project manager, make (reasoning that since you are charged with managing the project, you should also be the one to choose the architecture that makes that task the easiest), unfortunately that will not work in many cases. The choice of approach must take into consideration the uses to which the WBS will be put. What may be the best choice for defining the work to be done may not be the best choice for status reporting.

There is no one correct way to create the WBS. Hypothetically, if you put each member of the JPPS in a different room and ask them all to develop the project WBS, they might all come back with different renditions. That's all right — there is no single best answer. The choice is subjective and based more on the project manager's preference than on any other requirements. In practice, I have sometimes tried to follow one approach only to find that it was making the project work more confusing, rather than simpler. In such cases, my advice is simply to throw away the work you have done and start all over again with a fresh approach.

The three general approaches to building the WBS are as follows:

Noun-type approaches — These approaches define the deliverables of the project work in terms of the components (physical or functional) that make up the deliverable. These are the requirements that populate the RBS. This approach is the one currently recommended by PMI. If you have generated the RBS, you are very close to having a deliverables-based WBS. Figure 4-3 shows the relationship between the RBS and the WBS. First, note that the RBS is a subset of the WBS. To put it another way, the RBS defines what must be done, and the WBS defines how it will be done.

Verb-type approaches — These approaches define the deliverable of the project work in terms of the actions that must be done to produce that deliverable. Verb-type approaches include the

design-build-test-implement and project objectives approaches. This approach was recommended by PMI prior to the current release of PMBOK.

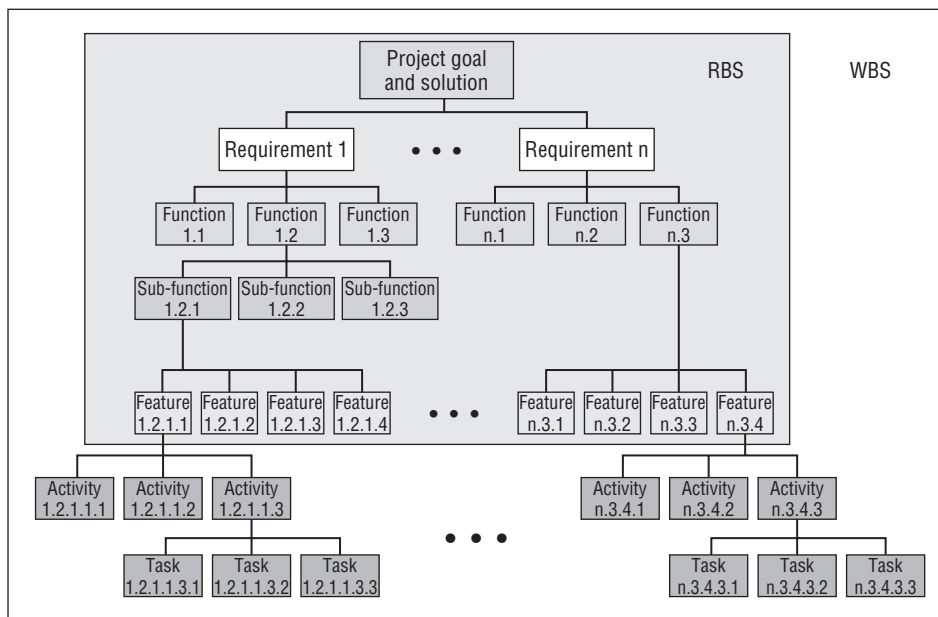


Figure 4-3 The relationship between the RBS and the WBS

Organizational approaches — These approaches define the deliverable of the project work in terms of the organizational units that will work on the project. This type of approach includes the department, process, and geographic location approaches.

You have probably seen one or more of these approaches used in practice to create the WBS. The following sections take a look at each of these approaches in more detail.

Noun-Type Approaches

There are two noun-type approaches: physical decomposition and functional decomposition.

Physical Decomposition

In projects that involve building products, it is tempting to follow the physical decomposition approach. Consider a mountain bike, for example. Its physical components include a frame, wheels, suspension, gears, and brakes. If each component is to be manufactured, this approach might produce a simple WBS.

As mentioned previously, though, you have to keep in mind the concern of summary reporting.

For example, consider rolling up all the tasks related to gears. If you were to create a Gantt chart for reporting at the summary level, the bar for the gears' summary activity would start at the project start date. A Gantt chart is a simple graphical representation of the work to be done and the schedule for completing it. The Gantt chart consists of a number of rectangular bars — each one representing an activity in the project. The length of each bar corresponds to the estimated time it will take to complete the activity. These bars are arranged across a horizontal time scale, with the left edge of the bar lined up with the scheduled start of the activity. The bars are arranged vertically in the order of scheduled start date. The resulting picture forms a descending stair-step pattern. That is, after all, where the detailed tasks of doing design work would occur. The finish of the bar would occur at about the project completion date. That is where testing and documentation for the gears occurs. Using the summary Gantt chart as a status reporting tool for the gears doesn't have much use. The bar extends from the beginning to the end of the timeline. The same is true of all the other physical components mentioned. Showing all of them on a summary Gantt chart would simply look like the stripes on a prison uniform.

This type of WBS is initially attractive because it looks similar to, and in fact could be identical to, a company's financial chart of accounts (CoA). A CoA is noun-oriented because it accounts for the cost of developing things such as gears and brakes. A CoA should not be confused with the WBS. The WBS is a breakdown of work; the CoA is a breakdown of costs. Most popular project-management software products provide code fields that can be used to link project task costs with accounting CoA categories.

Functional Decomposition

The WBS can also be built based on the functional components of the product. Using the mountain bike example from the preceding section, the functional components would include the steering system, gear-shifting system, braking system, and pedaling system. The same cautions that apply to the physical decomposition approach apply here as well.

Verb-Type Approaches

There are two verb-type approaches: the design-build-test-implement approach and the objectives approach.

Design-Build-Test-Implement

The design-build-test-implement approach is commonly used in projects that involve a methodology. Application systems development is an obvious

example. Using the bicycle example again, a variation on the classic waterfall categories could be used. The categories are design, build, test, and implement. If you use this architecture for your WBS, then the bars on the Gantt chart would all have lengths that correspond to the duration of each of the design, build, test, and implement activities, and hence would be shorter than the bar representing the entire project. Most, if not all, would have differing start and end dates. Arranged on the chart, they would cascade in a stair-step (“waterfall”) manner. These are just representative categories — yours may be different. The point is that when the detail-level activity schedules are summarized up to them, they present a display of meaningful information to the recipient of the report.

Remember, the WBS activities at the lowest levels of granularity must always be expressed in verb form. After all, this is *work*, which implies action, which in turn implies verbs.

Objectives

The objectives approach is similar to the design-build-test-implement approach and is used when progress reports at various stages of project completion are prepared for senior management. Reporting project completion by objectives provides a good indication of the deliverables that have been produced by the project team. Objectives are almost always related to business value and will be well received by senior management as well as the client. There is a caveat, however. This approach can cause some difficulty because objectives often overlap. Their boundaries can be fuzzy. If you use this approach, you’ll have to give more attention to eliminating redundancies and discovering gaps in the defined work.

Organizational Approaches

The deployment of project work across geographic or organizational boundaries often suggests a WBS that parallels the organization. The project manager would not choose to use this approach but rather would use it only when forced to because of the organizational structure. In other words, the project manager has no other reasonable choice. These approaches offer no real advantages and tend to create more problems than they solve. However, they are described here in case you have no other option for building the WBS.

Geographic

If project work is geographically dispersed, it may make sense from coordination and communication perspectives to partition the work first by geographic location and then by some other approach at each location. For example, a

project for the U.S. space program because of its geographic components might require this type of approach.

Departmental

Departmental boundaries and politics being what they are, you might benefit from partitioning the project first by department and then within each department by whatever approach makes sense. You benefit from this structure in that a major portion of the project work is under the organizational control of a single manager, which in turn simplifies resource allocation. Conversely, using this approach increases the need for communication and coordination across organizational boundaries.

Business Process

The final approach involves breaking the project down first by business process and then by some other method for each process. This has the same advantages and disadvantages as the departmental approach, with the added complication that integration of the deliverables from each process can be more difficult when you use this approach. The difficulty arises from process interactions at the boundaries of the involved processes. For example, at the boundary between order entry and order fulfillment how would order verification be defined? It could be part of either process. The process in which you place it will impact the customer.

Selecting the Best Approach

Again, no single approach can be judged to be best for a given project. My advice is to consider each approach at the outset of the JPPS and pick the one that seems to bring the most clarity to defining the project work.

Representing the WBS

Whatever approach you use, the WBS can be generically represented, as shown previously in Figure 4-2. The goal statement represents the reason for doing the project. At Level 1 partition the goal into some number of activities (also known as chunks of work). These chunks of work are a necessary and sufficient set that define the goal. That is, when all of these first-level activities are complete, the goal is met and the project is complete.

Partition any activity that does not possess the six characteristics into a set of necessary and sufficient activities at the next level of decomposition. The process continues until all activities have met the six criteria. The lowest level of decomposition in the WBS defines a set of activities (renamed “tasks”) that

will each have a task manager, someone who is responsible for completing the task.

Tasks are further defined by a work package. A work package is simply the list of things to do to complete the task. The work package may be very simple, such as getting management to sign off on a deliverable, or it may be a mini-project and consist of all the properties of any other project, except that the activity defining this project possesses the six criteria and need not be further partitioned.

CROSS-REF Chapter 5 describes how to create and use work packages.

Some examples will help clarify this idea. Figure 4-4 is a partial WBS for building a house, and Figure 4-5 is the indented outline version (for those of you who prefer an outline format to a hierarchical graph). Both convey the same information.

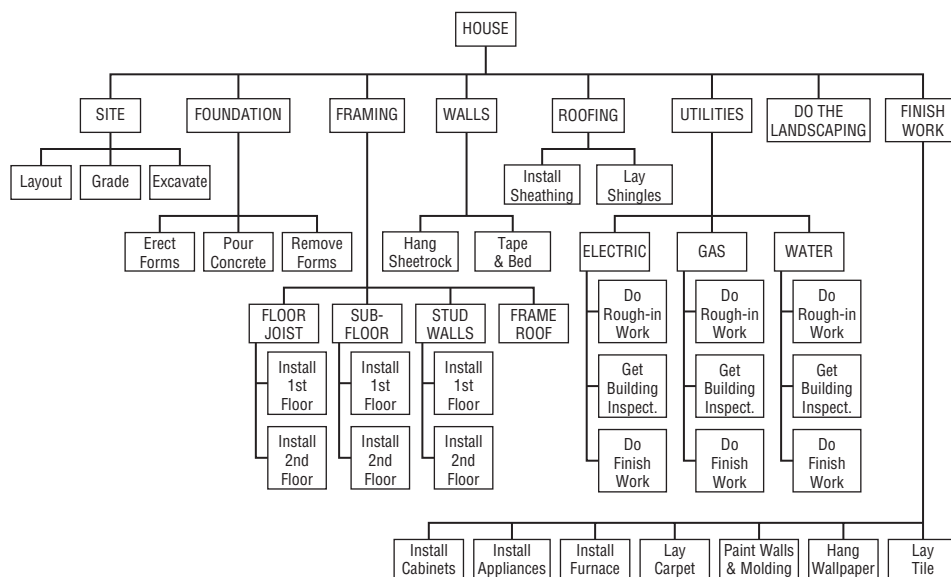


Figure 4-4 WBS for a house

Figure 4-6 shows the WBS for the traditional waterfall systems development methodology. If you're a systems project manager, this format could become a template for all your systems development projects. It is a good way to introduce standardization into your systems development methodology.

-
- 1. SITE PREPARATION
 - 1.1 Layout
 - 1.2 Grading
 - 1.3 Excavation
 - 2. FOUNDATION
 - 2.1 Erect Forms
 - 2.2 Pour Concrete
 - 2.3 Remove Forms
 - 3. FRAMING
 - 3.1 Floor Joists
 - 3.1.1. Install First-Floor Joists
 - 3.1.2. Install Second-Floor Joists
 - 3.2 Subflooring
 - 3.2.1. Install First-Floor Subflooring
 - 3.2.2. Install Second-Floor Subflooring
 - 3.3 Stud Walls
 - 3.3.1. Erect First-Floor Stud Walls
 - 3.3.2. Erect Second-Floor Stud Walls
 - 3.4 Frame the Roof
 - 4. UTILITIES
 - 4.1 Electrical
 - 4.1.1. Do Rough-in Work
 - 4.1.2. Get Building Inspection
 - 4.1.3. Do Finish Work
 - 4.2 Gas
 - 4.2.1. Do Rough-in Work
 - 4.2.2. Get Building Inspection
 - 4.2.3. Do Finish Work
 - 4.3 Water
 - 4.3.1. Do Rough-in Work
 - 4.3.2. Get Building Inspection
 - 4.3.3. Do Finish Work
 - 5. WALLS
 - 5.1 Hang Sheetrock
 - 5.2 Tape and Bed
 - 6. ROOFING
 - 6.1 Install Sheathing
 - 6.2 Lay Shingles
 - 7. FINISH WORK
 - 7.1 Install Cabinets
 - 7.2 Install Appliances
 - 7.3 Install Furnace
 - 7.4 Lay Carpet
 - 7.5 Paint Walls and Molding
 - 7.6 Hang Wallpaper
 - 7.7 Lay Tile
 - 8. LANDSCAPING

Figure 4-5 Indented outline WBS for a house

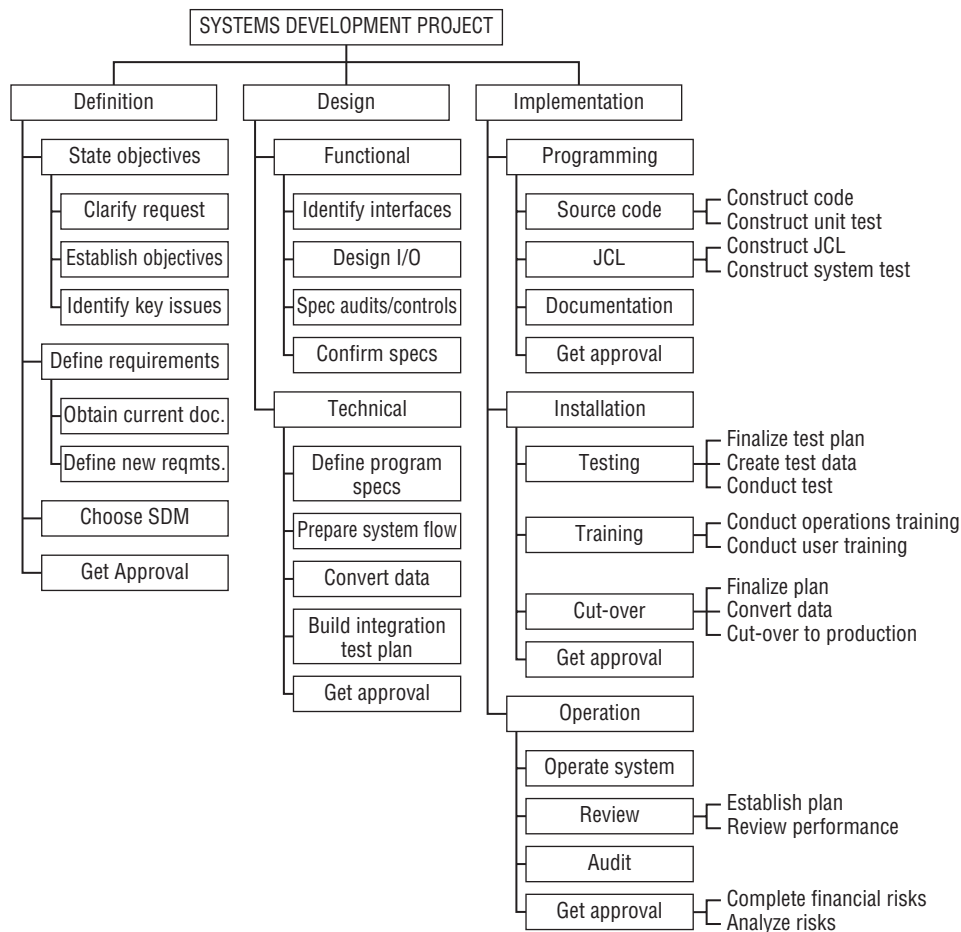


Figure 4-6 WBS for a waterfall systems development methodology

Estimating

Estimation is the one area where most project teams have trouble. For one thing, there is no consistency. One person might be optimistic, another pessimistic, and you won't know which unless you have had prior substantiating evidence of one or the other. Having the professional who will be responsible for the work also estimate the duration or labor involved is a good idea, but it isn't the answer either. Will this person be pessimistic just so the work is sure to meet the estimated deadline? The approach that I use is to have more than one person provide each estimate, as explained in this section.

Estimating Duration

Before you can estimate duration, you need to make sure everyone is working from a common definition. The duration of a project is the elapsed time in business working days, not including weekends, holidays, or other non-work days. Work effort is labor required to complete a task. That labor can be consecutive or nonconsecutive hours.

Duration and work effort are not the same thing. For example, I had a client pose the following situation: The client had a task that required him to send a document to his attorney, where it would be reviewed, marked up, and then returned. He had done this on several previous occasions, and it normally took about 10 business days before the document was back in his office. He knew the attorney took only about 30 minutes to review and mark up the document. His question was, “What’s the duration?” The answer is 10 days. The work effort on the part of the attorney is 30 minutes.

It is important to understand the difference between labor time and duration time. They are not the same. Suppose an estimate has been provided that a certain task requires 10 hours of focused and uninterrupted labor to complete. Under normal working conditions, how many hours do you think it will really take? Something more than 10 for sure. To see why this is so, consider the data shown in Figure 4-7.

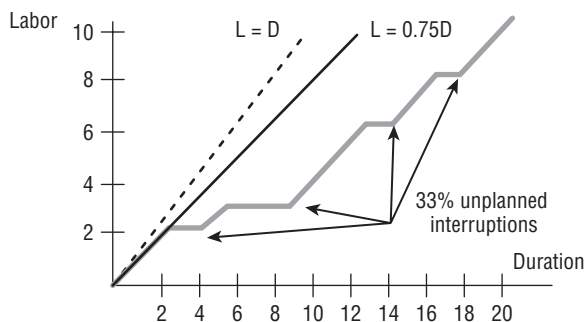


Figure 4-7 Elapsed time versus work time

If a person could be focused 100 percent of the time on a task, he or she could accomplish 10 hours of work in 10 hours. Such a person would indeed be unique, for it is more likely that his or her work will be interrupted by e-mail, cell phones or text messages, meetings, coffee breaks, and socializing. Several estimates have been made regarding the percentage of a person’s day that he or she can devote to project work. Past data that I have collected from information technology (IT) professionals indicates a range of 66 to 75 percent. More recently, among the same client base, I have seen a downward trend in

this estimate to a range of 50 to 65 percent. If you use the 75 percent estimate, a 10-hour task should require about 13 hours and 20 minutes to complete. That is *without* interruptions, which, of course, always happen.

For some professionals (those who provide technical support, for example), interruptions are frequent; for others (such as testing staff), interruptions are infrequent. I polled the 17-person technical support unit of a midsize IT department and found that about one-third of their time was spent on unplanned interruptions. Unplanned interruptions might include a phone call with a question to be answered, a systems crash, power interrupts, random events of nature, the boss stopping in to discuss an unrelated matter, or a personal call from your golfing buddies. Using a 75 percent estimate for focused work and a 33 percent estimate for unplanned interruptions, a 10-hour task will take approximately 20 hours to complete.

It is this elapsed time that you are interested in when estimating for each task in the project. It is the true duration of the task. For costing purposes, you are interested in the labor time (work) actually spent on a task.

NOTE When estimating task duration, you have a choice to make: Do you want to estimate hours of billable labor to complete the task, or do you want to estimate the clock time required to complete the task? You will probably want to do both. The labor hours are needed in order to bill the client. The elapsed clock time is needed to estimate the project completion date. Some project managers will estimate labor and convert it to duration by dividing labor time by an established efficiency factor, typically ranging from 0.6 to 0.75.

Resource Loading versus Task Duration

The duration of a task is influenced by the amount of resources scheduled to work on it. I say “influenced” because there is not necessarily a direct linear relationship between the amount of resources assigned to a task and its duration.

Adding more resources to hold a task’s duration within the planning limits can be effective. This is called “crashing the task.” For example, suppose you are in a room where an ordinary-size, four-legged chair is in the way. The door to the room is closed. You are asked to pick up the chair and take it out of the room into the hallway. You might try to do it without any help, in which case you would perform the following steps:

1. Pick up the chair.
2. Carry it to the door.
3. Set the chair down.
4. Open the door.

5. Hold the door open with your foot as you pick up the chair.
6. Carry the chair through the door.
7. Set the chair down in the hallway.

Suppose you double the resources. You'll get someone to help you by opening the door and holding it open while you pick up the chair and carry it out to the hallway. With two people working on the task, you'd probably be willing to say it would reduce the time needed to move the chair out of the room and into the hallway.

Doubling the resources sounds like a technology breakthrough in shortening duration. Now double them again and see what happens. Now you've got four resources assigned to the task. It would go something like this: First, you hold a committee meeting to decide roles and responsibilities. Who's in charge? Who holds the door open? Who takes what part of the chair? The duration actually increases!

The point of this silly example is to demonstrate that there may be diminishing returns for adding more resources. You would probably agree that there is a maximum loading of resources on a task to minimize the task duration, and that by adding another resource, you will actually begin to increase the duration. You have reached the crashpoint of the task. The *crashpoint* is the point where adding more resources will increase task duration. The project manager will frequently have to consider the optimum loading of a resource on a task.

A second consideration for the project manager is the amount of reduction in duration that results from adding resources. The relationship is not linear. Consider the chair example again. Does doubling the resources cut the duration in half? Can two people dig a hole twice as fast as one? Probably not. The explanation is simple. By adding the n th person to a task, you create the need for n more communication links. Who is going to do what? How can the work of several people be coordinated? There may be other considerations that actually add work. To assume that the amount of work remains constant as you add resources is simply not correct. New kinds of work will emerge from the addition of a resource to a task. For example, adding another person adds the need to communicate with more people, which increases the duration of the task.

A third consideration for the project manager is the impact on risk that results from adding another resource. If you limit the resource to people, you must consider the possibility that two people will prefer to approach the task in different ways, with different work habits and different levels of commitment. The more people working on a task, the more likely one will be absent, the higher the likelihood of a mistake being made, and the more likely they will get in each other's way.

The fourth consideration has to do with partitioning the task so that more than one resource can work on it simultaneously. For some tasks, this will be easy; for others it may be impossible. For example, painting a house is a partitionable task. Rooms can be done by different painters, and even each wall can be done by a different painters. The point of diminishing returns is not an issue here. Conversely, the task of writing a computer program may not be partitionable at all. Adding a second programmer creates all kinds of work that wasn't present with a single programmer — for example, choosing a language and/or naming conventions to use, integration testing, and so on.

Variation in Task Duration

Task duration is a random variable. Because you cannot know what factors will be operative when work is underway on a task, you cannot know exactly how long it will take. There will, of course, be varying estimates with varying precision for each task. One of your goals in estimating task duration is to define the task to a level of granularity such that your estimates have a narrow variance — that is, the estimate is as good as you can get it at the planning stages of the project. As project work is completed, you will be able to improve the earlier estimates of activities scheduled later in the project.

The following factors can cause variation in the actual task duration:

Varying skill levels — Your strategy is to estimate task duration based on using people of average skills assigned to work on the task. In actuality, this may not happen. You may get a higher- or lower-skilled person assigned to the task, causing the actual duration to vary from planned duration. These varying skill levels will be both a help and a hindrance to you.

Unexpected events — Murphy's Law is lurking around every bend in the road and will surely make his presence known, but in what way and at what time you do not know. Random acts of nature, vendor delays, incorrect shipments of materials, traffic jams, power failures, and sabotage are but a few of the possibilities.

Efficiency of worker's time — Every time a worker is interrupted, it takes additional time to get back to the level of productivity attained prior to the interruption. You cannot control the frequency or time of interruptions, but you do know that they will happen. As to their effect on staff productivity, you can only guess. Some will be more affected than others.

Mistakes and misunderstandings — Despite all of your efforts to clearly and concisely describe each task that is to be performed, you will most likely miss a few. This will take its toll in rework or scrapping semicompleted work.

Common cause variation — A task's duration will vary simply because *duration* is a random variable. The process has a natural variation, and there is nothing you do can to cause a favorable change in that variation. It is there and must be accepted.

Six Methods for Estimating Task Duration

Estimating task duration is challenging. You can be on familiar ground for some activities and on totally unfamiliar ground for others. Whatever the case, you must produce an estimate. It is important that senior management understand that the estimate can be little more than a WAG (wild a** guess). In many projects, the estimate will improve as you learn more about the deliverables after having completed some of the project work. Re-estimation and replanning are common. In my consulting practice, I have found the following six techniques to be quite suitable for initial planning estimates:

- Similarity to other activities
- Historical data
- Expert advice
- Delphi technique
- Three-point technique
- Wide-band Delphi technique

The following sections describe each of these techniques in more detail.

Extrapolating Based on Similarity to Other Activities

Some of the activities in your WBS may be similar to activities completed in other projects. Your or others' recollections of those activities and their durations can be used to estimate the present task's duration. In some cases, this process may require extrapolating from the other task to this one. In most cases, using the estimates from those activities provides estimates that are good enough.

Studying Historical Data

Every good project management methodology includes a project notebook that records the estimated and actual task durations. This historical record can be used on other projects. The recorded data becomes your knowledge base for estimating task duration. This technique differs from the previous technique in that it uses a record, rather than depending on memory. To estimate the

duration of a task, you extract similar tasks from the database and compute an average. That is a simple application of the database.

The historical data can be used in a more sophisticated way, too. One of my clients built an extensive database of task duration history. They use this database to record not only estimated and actual duration, but also the characteristics of the task, the skill set of the people working on it, and other variables that they found useful. When a task duration estimate is needed, they go to their database with a complete definition of the task and, with some rather sophisticated regression models, estimate the task duration. This particular client builds product for market, so it is very important to them to be able to estimate as accurately as possible. Again, my advice is that if there is value added for a particular tool or technique, use it.

Seeking Expert Advice

When the project involves a breakthrough technology or a technology that is being used for the first time in the organization, there may not be any organizational experience with that technology within the organization. In these cases, you will have to appeal to outside authorities. Vendors may be a good source, as are non-competitors who use that technology.

Applying the Delphi Technique

The Delphi technique can produce good estimates in the absence of expert advice. This is a group technique that extracts and summarizes the knowledge of the group to arrive at an estimate. After the group is briefed on the project and the nature of the task, each individual in the group is asked to make his or her best guess of the task duration. The results are tabulated and presented to the group in a histogram labeled First Pass, as shown in Figure 4-8. Participants whose estimates fall in the outer quartiles are asked to share the reason for their guess. After listening to the arguments, each group member is asked to guess again. The results are presented as a histogram labeled Second Pass, and again the outer quartile estimates are defended. A third guess is made, and the histogram plotted is labeled Third Pass. Final adjustments are allowed. The average of the third guess is used as the group's estimate. Even though the technique seems rather simplistic, it has been shown to be effective in the absence of expert advice.

I attended an IBM business partners' meeting several years ago. One of the sessions dealt with estimating software development time, and the presenter demonstrated the use of the Delphi technique with a rather intriguing example. She asked if anyone in the group had ever worked in a carnival as a weight-guessing expert. None had, so she informed the group that they were going to use the Delphi technique to estimate the average weight of the 20

people who were in the room. She asked everyone to write his or her weight on a slip of paper. The weights were averaged by the facilitator and put aside. Each person took an initial guess as to the average weight of the people in the room, wrote it down, and passed it to the facilitator. She displayed the initial-pass histogram and asked the individuals with the five highest and five lowest guesses to share their thinking with the group; a second guess was taken and then a third. The average of the third guess became the group's estimate of the average body weight. Surprisingly, the estimate was just two pounds off the average of the individual's reported weights.

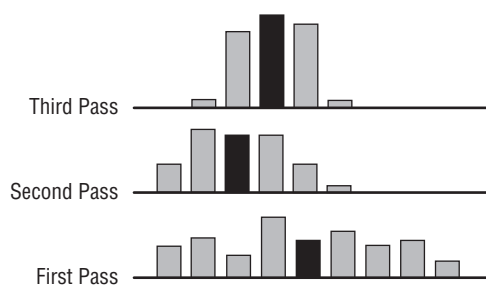


Figure 4-8 The Delphi technique

The approach the presenter used is actually a variation of the original Delphi technique. The original version used a small panel of experts (say, five or six) who were asked for their estimate independently of one another. The results were tabulated and shared with the panel members, who were then asked to submit their own second estimate. A third estimate was solicited in the same manner. The average of the third estimate was the one chosen. Note that the original approach does not involve any discussion or collaboration between the panel members. In fact, they weren't even aware of who the other members were.

Applying the Three-Point Technique

Task duration is a random variable. If it were possible to repeat the task several times under identical circumstances, duration times would vary. That variation may be tightly grouped around a central value, or it might be widely dispersed. In the first case, you would have a considerable amount of information on that task's duration as compared to the latter case, where you would have very little or no usable information on that task's duration. In any given instance of the task, you would not know at which extreme the duration would likely fall, but you could make probabilistic statements about their likelihood in any case. Figure 4-9 illustrates the point.

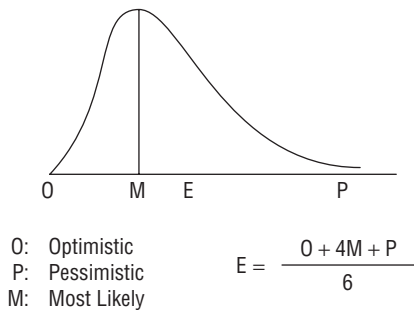


Figure 4-9 The three-point technique

To use the three-point technique, you need the following three estimates of task duration:

Optimistic — The optimistic time is defined as the shortest duration one has experienced or might expect to experience given that everything happens as expected.

Pessimistic — The pessimistic time is that duration that one would experience (or has experienced) if everything that could go wrong did go wrong, yet the task was completed.

Most likely — The most likely time is that time that they usually experience.

For this method, you are calling on the collective memory of professionals who have worked on similar activities but for which there is no recorded history.

Applying the Wide-Band Delphi Technique

Combining the Delphi and three-point methods results in the wide-band Delphi technique. It involves a panel, as in the Delphi method. In place of a single estimate, the panel members are asked, at each iteration, to give their optimistic, pessimistic, and most likely estimates for the duration of the chosen task. The results are compiled, and any extreme estimates are removed. Averages are computed for each of the three estimates, and the averages are used as the optimistic, pessimistic, and most likely estimates of task duration.

Estimation Life Cycles

A word of advice on estimating is in order here. Early estimates of task duration will not be as good as later estimates. It's a simple fact that you get smarter as the project work commences. Estimates will always be subject to

the vagaries of nature and other unforeseen events. You can only hope that you gain some knowledge through the project to improve your estimates.

In the top-down project planning model, you start out with “roughly right” estimates, with the intention of improving the precision of these estimates later in the project. Both your upper management and the client must be aware of this approach. Most managers have the habit of assuming that a number, once written, is inviolate and absolutely correct regardless of the circumstances under which the number was determined. This is unrealistic in the contemporary world of project management. It may have once been appropriate for the engineer, but that is not the case with the businessperson. Figure 4-10 illustrates a typical estimation life cycle.

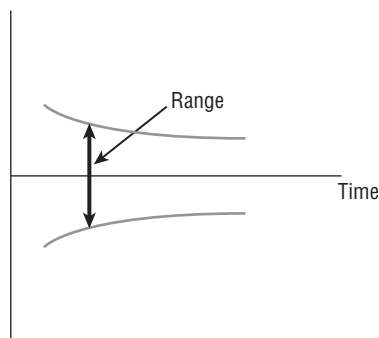


Figure 4-10 The estimation life cycle

During project planning, most task estimates are not much better than a dart throw. Of course, if the task is one that has been done several times before under similar situations, then the estimate will have a narrower variance than it would if the task had never been done before under any circumstances. As project work commences, the team will gain knowledge and understanding of the project and the work needed to deliver an acceptable solution. That includes improved estimates of future tasks in the project. After the task has begun, even more information will come to light, and the estimate (or a better estimate to completion) will be quite accurate.

Estimating Resource Requirements

By defining project activities according to the completion criteria, you should have reached a point of granularity with each task so that it is familiar. You may have done the task or something very similar to it in a past project. That recollection, or historical information, gives you the basis for estimating the resources you will need to complete the activities in the current project. In some cases, it is a straightforward recollection. In others, it is the result of

keeping a historical file of similar activities. In still others, the resource estimate may be the advice of experts.

The importance of resources varies from project to project. You can use the six techniques discussed in the previous section to estimate the resource requirements for any project.

Types of resources include the following:

People — In most cases, the resources you will have to schedule are human resources. This is also the most difficult type of resource to schedule.

Facilities — Project work takes place in locations. Planning rooms, conference rooms, presentation rooms, and auditoriums are but a few examples of facilities that projects require. The exact specifications of the required facilities as well as the precise time at which each facility is needed are some of the variables that you must take into account. The project plan can provide the required details. The availability of the facilities will also drive the project schedule.

Equipment — Equipment is treated exactly the same as facilities. The availability of equipment will also drive the task schedule.

Money — Accountants will tell you that everything is eventually reduced to dollars, which is true. Project expenses typically include travel, accommodations, meals, and supplies.

Materials — The timely availability of parts to be used in the fabrication of products and other physical deliverables will be part of the project work schedule. For example, the materials needed to build a bicycle might include nuts, bolts, washers, and spacers.

People as Resources

People are the most difficult type of resource to schedule because you plan the project by specifying the types of skills you need, when you need them, and in what amounts. You do not specify the resource by name (that is, the individual you need), which is where problems arise.

There are a few tools you can use to help you schedule people.

Skills Matrices

I find that an increasing number of my clients are developing skills-inventory matrices for staff, and skill-needs matrices for activities. The two matrices are used to assign staff to activities. The assignment could be based on task characteristics such as risk, business value, criticality, and/or skill development. Figure 4-11 illustrates how the process can work.

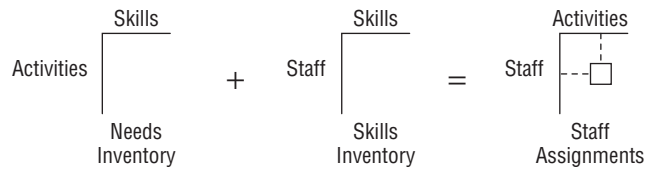


Figure 4-11 Assigning staff to activities

This process involves gathering data for the following two inventories:

- An inventory of the demand for skills needed to perform the tasks associated with specific activities. This is represented as a matrix whose rows are the activities and whose columns are the skills. These include both current and long-term needs.
- An inventory of the current skills among the professional staff. This is represented as a matrix whose rows identify the staff and whose columns represent the skills.

The columns of both matrices define the same set of skills. This gives you a way to link the two matrices and assign staff to activities. This approach can be used for on-the-job staff development. As an on-the-job development strategy, the manager would have previously met with the staff member, helped him or her define career goals, and translated those goals into skill development needs. That information can then be used in project planning to assign staff to activities so that the work they will do on the task enables them to meet those goals.

Skill Categories

This part of the skill matrix is developed by looking at each task that the unit must perform and describing the skills needed to perform the task. Because skills may appear in unrelated activities, the list of possible skills must be standardized across the enterprise.

Skill Levels

A binary assessment that simply determines whether a person has the necessary skill or doesn't is certainly easier to administer, but it isn't sufficient for project management. Skills must be qualified with a statement about how much of the skill the person possesses. Various methods are available, and companies often develop their own skill-level system.

Resource Breakdown Structure

Just as there is a Work Breakdown Structure (WBS), so also is there a Resource Breakdown Structure (RBS). Figure 4-12 gives a simple example.

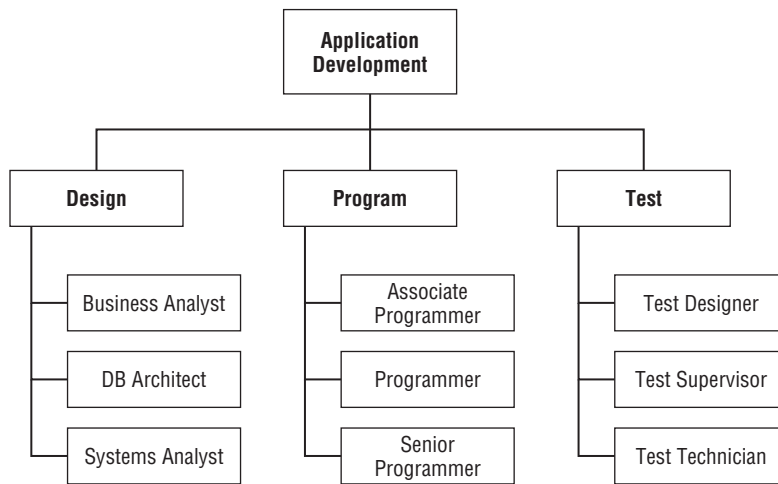


Figure 4-12 Example of a Resource Breakdown Structure

The RBS is used to assist in not only resource estimation but also cost estimation. The RBS is determined by the job families, which are defined by the human resources department. That definition is simply put into this hierarchical framework and used as the basis for identifying the positions and levels that are needed to staff the project. This is then used to construct the staffing budget.

Determining Resource Requirements

The planning team includes resource managers or their representatives. At the time the planning team is defining the WBS and estimating task duration, they also estimate resource requirements.

I have found the following practice effective:

1. Create a list of all the resources required for the project. For human resources, list only the position title or skill level. Do not name specific people even if there is only one person with the requisite skills. Envision a person with the typical skill set and loading on the project task. Task duration estimates are based on using workers of average skill level, and that should be consistent with the needed resource requirements. You can worry about changing the assumption of using average skilled persons later in the planning session.
2. The project manager can provide, resource requirements as part of the WBS.

You now have estimated the parameters needed to begin constructing the project schedule. The task duration estimates provide input to planning the

order and sequence of completing the work defined by the activities. After the initial schedule is built, you can use the resource requirements and availability data to further modify the schedule.

Resource Planning

You need to consider several factors concerning the resources for your project. As you learned earlier in this chapter, adding resources doesn't necessarily mean that you will shorten the time needed for various activities. Adding too many people can actually add time to the project. Another factor to consider deals with the skill level of the resources.

Suppose that you are going to have a team of developers work on an application. When you are planning resources, you have to know the skill level of the potential resources. You may have to trade money for time. That means you may be able to get a lower cost by using a junior developer, but it's likely you will also find that the process takes longer. Knowing the skill sets of the available people and taking that into account when doing scheduling are critical to resource planning.

Another factor to consider is that of using existing staff on a part-time basis. At first glance, using staff on a part-time basis might seem like a good idea because you can make good decisions concerning their schedules and you will get extremely efficient use of their time. However, this isn't reality, particularly if they are coders. Development is a mental task, and what you need are knowledge workers. You can't turn a mental process on and off at will.

Similarly, scheduling people to work on two projects on the same day won't be an efficient use of resources. It takes some time for a developer to get up to writing speed. That kind of schedule may look good on paper, but it doesn't work. Give your resources a chance to get into the flow of work, and you'll be much more successful.

WHAT IF THE SPECIFIC RESOURCE IS KNOWN?

Knowing the specific resource will occur quite often, and when it does, you will be faced with some questions. Should you put that person in the plan? If you do and that person is not available when you need him or her, how will that affect your project plan? If he or she is very highly skilled and you used that information to estimate the duration of the task that person was to work on, you may have a problem. If you cannot replace him or her with an equally skilled individual, will that create a slippage that has a domino effect on the project schedule? Make your choice.

Estimating Cost

After you have estimated task duration and resource requirements, you have the data you need to establish the cost of the project. This is your first look at the dollars involved in doing the project. You know the resources that will be required and the number of hours or volume of resources needed. You can now estimate the project cost by applying the unit cost data to the amount of resources required.

When doing an estimate, you need to consider a few concepts. No matter how well you estimate cost, it is always an estimate. One of the reasons that so many projects come in over budget is that people actually believe that they have done perfect estimating and that their baseline estimate is set in stone. Remember that it is still and always will be an estimate. Anytime you are forecasting the future, as you are when you plan a project, you are dealing with some amount of uncertainty. Projects are so often over budget because the budget itself is an estimate, not an exact mathematical calculation. Even experienced cost estimators miss the mark.

With those warnings in mind, you still need to do your best to come up with a working budget for the project. Estimating can be done several ways. One method is to find an analogous project — a completed project that looks a great deal like the project you're currently planning. By using this previous project as a guideline, you'll have a reference point for the costs. The caveat that you must keep in mind is that each project is unique, which means you will have a slightly different budget for your estimate than the one from the previous project. Don't just use the same figures when you estimate, or this will come back to haunt you sometime in the project.

Another good practice in estimating is to invite subject matter experts (SMEs) to help you prepare your estimate. Ideally, these SMEs will be able to discuss their areas of expertise and give you a better handle on the estimating for your current projects.

The team should have access to a standard costing table. This table will list all resources, units of measure, and cost per unit. It is then just a simple exercise in calculating the cost per resource based on the number of units required and the cost per unit. Many organizations have a spreadsheet template that will facilitate the exercise. These calculated figures can be transferred to the WBS and aggregated up the WBS hierarchy to provide a total cost for each task level in the WBS.

The following three types of estimates are common in project management. They are often done in the sequence listed.

Order of magnitude estimate — This type means that the number given for the estimate is somewhere between 25 percent above and 75 percent below the number. Order of magnitude estimates are often used at the very beginning of the estimation process when very little detail is known about the project work and a rough estimate is all that management calls for. It is understood that this estimate will be improved over time. This estimate is often a very preliminary one and is done just to get some sense as to the financial feasibility of the project.

Budget estimate — This type can typically have a range of 10 percent over and 25 percent below the stated estimate. These estimates are generated during project planning and are based on knowing some detail about the project activities.

Definitive estimate. This type is generally the one that is used for the rest of the project. It has a range of 5 percent over and 10 percent below the stated estimate. These estimates are done frequently during project execution when new information helps further improve the range of the estimates generated during project planning.

NOTE When giving an estimate for a project, it's a good idea to have the three ranges discussed above in mind. Remember that even if you tell your client that your estimate is an order of magnitude estimate, they will not remember it for the rest of the project. Protect yourself in this case by writing "Order of Magnitude" on the estimate. Doing so will at least give you something with which to defend yourself if it comes to that. Make it a point to let all concerned parties know about the different types of estimates you are providing and how those estimates are used.

Cost Budgeting

After you've done an estimate, you enter the cost budgeting phase. This is the phase when you assign costs to tasks on the WBS. Cost budgeting is actually very formulaic. You take the needed resources and multiply the costs by the number of hours they are to be used. In the case of a one-time cost (such as hardware), you simply state that cost.

Cost budgeting gives the sponsor a final check on the costs of the project. The underlying assumption is that you've got all the numbers right. Usually you'll have the cost of a resource right, but often it's tough to be exact on the total number of hours the resource is to be used. Remember that no matter what, you are still doing an estimate. Cost budgeting is different from estimating

in that it is more detailed. However, the final output is still a best effort at expressing the cost of the project.

Cost Control

Cost control presents the following major issues for the project manager:

- How often do you need to get reports of the costs? Certainly, it would be good if you could account for everything occurring on the project in real time. However, that is generally way too expensive and time-intensive. More likely, you'll get figures once a week. Getting cost status figures once a week gives you a good snapshot of the costs that are occurring. If you wait longer than a week to get cost figures, you may find that the project has spun out of control.
- How will you look at the numbers you're receiving? If you've done a cost baseline, then you'll have some figures against which you can measure your costs. What you're looking for is a variance from the original costs. The two costs you have at this point are your baseline and the actual costs that have occurred on the project. The baseline was the final estimate of the costs on the project. Your job is to look at the two and determine whether management action must be taken.

TIP How far off do the numbers need to be between the final estimate and your actual costs before you take some action? Usually 10 percent is the most allowed. However, if you start to see a trend of the plan going over budget or behind schedule or both — you should take a look at the reasons behind the variances before they reach the 10 percent level. See Chapter 6 for more details.

A word of advice: Keep in mind that for some projects, time is the most important constraint. Remember Y2K? In such cases, you must balance your need for cost control with the need for a definitive project ending time. There may be a trade-off in cost control for time constraints. As a project manager, you must be aware of these trade-offs and be ready to justify changes in costs to the sponsor based on other considerations such as time and quality.

Constructing the Project Network Diagram

At this point in the PMLC, you have identified the known set of tasks in the project as output from building the RBS and WBS as well as the task duration for the project. Next, the planning team needs to determine the order in which these tasks are to be performed.

The tasks and the task duration are the basic building blocks needed to construct a graphic picture of the project. This graphic picture provides you with the following two additional pieces of schedule information about the project:

- The earliest time at which work can begin on every task that makes up the project
- The earliest expected completion date of the project

This is critical information for the project manager. Ideally, the required resources will be available at the times established in the project plan. This is not very likely, and Chapter 5 discusses how to deal with that problem. But first, you need to know how to create an initial project network diagram and the associated project schedule, which is the focus of this section.

Envisioning a Complex Project Network Diagram

A *project network diagram* is a pictorial representation of the sequence in which the project work can be done. You need to follow a few simple rules to build the project network diagram.

Recall from Chapter 1 that a project is defined as a sequence of interconnected tasks. You could simply perform the tasks one at a time until they are all complete, but in most projects, this approach would not result in an acceptable completion date. In fact, it results in the longest time to complete the project. Any ordering that allows even one pair of tasks to be worked on concurrently results in a shorter project completion date.

Another approach is to establish a network of relationships between the tasks. You can do this by looking forward through the project. What tasks must be complete before another task can begin? Conversely, you can take a set of tasks and look backward through the project: Now that a set of tasks is complete, what task or tasks could come next? Both methods are valid. The one you use is a matter of personal preference. Are you more comfortable looking backward in time or forward? My advice is to look at the tasks from both angles. One can be a check of the completeness of the other.

The relationships between the tasks in the project are represented in a flow diagram called a *network diagram* or *logic diagram*.

Benefits to Network-Based Scheduling

A project schedule can be built using either of the following:

- Gantt chart
- Network diagram

The Gantt chart is the oldest of the two and is used effectively in simple, short-duration types of projects. To build a Gantt chart, you begin by associating a rectangular bar with every task. The length of the bar corresponds to the duration of the task. You then place the bars horizontally along a timeline in the order in which the tasks should be completed. In some instances you will be able to schedule and work on tasks concurrently. The sequencing is often driven more by resource availability than any other consideration.

There are two drawbacks to using the Gantt chart. They are as follows:

- Because of its simplicity, the Gantt chart does not contain detailed information. It reflects only the order imposed by the manager and, in fact, hides much of that information. In other words, the Gantt chart does not contain all of the sequencing information that exists. Unless you are intimately familiar with the project tasks, you cannot tell from the Gantt chart what must come before and after what.
- The Gantt chart does not tell the project manager whether the schedule that results from the chart completes the project in the shortest possible time or even uses the resources most effectively. The Gantt chart reflects only when the manager would like to have the work done.

Although a Gantt chart is easier to build and does not require the use of an automated tool, I recommend using the network diagram. The network diagram provides a visual layout of the sequence in which project work flows. It includes detailed information and serves as an analytical tool for project scheduling and resource management problems as they arise during the life of the project. In addition, the network diagram enables you to compute the earliest time at which the project can be completed. That information does not follow from a Gantt chart.

Network diagrams can be used for detailed project planning, during implementation as a tool for analyzing scheduling alternatives, and as a control tool as described below.

Planning — Even for large projects, the project network diagram gives a clear graphical picture of the relationship between project tasks. It is, at the same time, a high-level and detailed-level view of the project. I have found that displaying the network diagram on the whiteboard or flip charts during the planning phase is beneficial. This way, all members of the planning team can use it for scheduling decisions.

Implementation — If you are using automated project management software tools, update the project file with task status and estimate-to-completion data. The network diagram is then automatically updated and can be printed or viewed. The need for rescheduling and resource reallocation decisions can be determined from the network diagram, although some argue that this

method is too cumbersome because of project size. Even a project of modest size (such as one that involves about 100 tasks) produces a network diagram that is too large and awkward to be of much use. I cannot disagree, but I place the onus on software manufacturers to market products that do a better job of displaying network diagrams.

Control — Although the updated network diagram retains the status of all tasks, the best graphical report for monitoring and controlling project work will be the Gantt chart view of the network diagram. This Gantt chart cannot be used for control purposes unless you have done network scheduling or incorporated the logic into the Gantt chart. Comparing the planned schedule with the actual schedule, you can discover variances and, depending on their severity, be able to put a get-well plan in place.

CROSS-REF Chapter 6 examines progress monitoring and control in more detail and describes additional reporting tools for analyzing project status.

Building the Network Diagram Using the Precedence Diagramming Method

One early method for representing project tasks as a network dates back to the early 1950s and the Polaris Missile Program. It is called the *task-on-the-arrow (TOA) method*. As Figure 4-13 shows, an arrow represents each task. The node at the left edge of the arrow is the event that begins the task, and the node at the right edge of the arrow is the event that ends the task. Every task is represented by this configuration. Nodes are numbered sequentially, and in the early versions of this method, the sequential ordering had to be preserved. Because of the limitations of the TOA method, ghost tasks had to be added to preserve network integrity. Only the simplest of dependency relationships could be used. This method proved to be quite cumbersome as networking techniques progressed. This approach is seldom used these days.

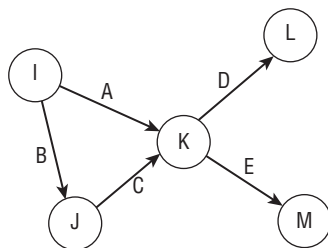


Figure 4-13 The TOA method

With the advent of the computer, the TOA method lost its appeal, and a new method replaced it. This method is called the *task-on-the-node (TON) method*, or more commonly, the *precedence diagramming method (PDM)*.

The basic *unit of analysis* in a network diagram is the task. Each task in the network diagram is represented by a rectangle called a task *node*. Arrows represent the predecessor/successor relationships between tasks. Figure 4-14 shows an example of a project network diagram in PDM format. You will take a more detailed look at how the PDM works later in this chapter.

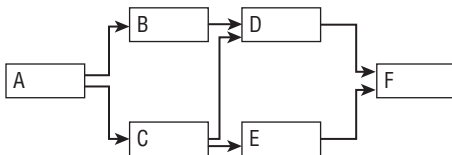


Figure 4-14 PDM format of a project network diagram

Every task in the project will have its own task node (see Figure 4-15). The entries in the task node describe the time-related properties of the task. Some of the entries describe characteristics of the task, such as its expected duration (E), whereas others describe calculated values (ES, EF, LS, and LF) associated with that task. (These terms will be defined shortly, and you'll see examples of their use.)

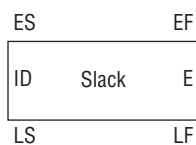


Figure 4-15 Task node

In order to create the network diagram using the PDM, you need to determine the predecessors and successors for each task. To do this, you ask: "What tasks must be complete before I can begin this task?" Here, you are looking for the technical dependencies between tasks. After a task is complete, it will have produced an output, which is a deliverable that becomes input to its successor tasks. Work on the successor tasks requires only the output from its predecessor tasks.

NOTE Later, you can incorporate management constraints that may alter these dependency relationships. In my experience, considering these constraints at this point in project planning only complicates the process.

What is the next step? Although the list of predecessors and successors to each task contains all the information you need to proceed with the project, it does not represent the information in a format that tells the story of your project. Your goal is to provide a graphical picture of the project. To do that, you need to understand a few rules. When you know the rules, you can create the graphical image of the project. This section teaches you the simple rules for constructing a project network diagram.

The network diagram is logically sequenced to be read from left to right. With the exception of the start and end tasks, every task in the network must have at least one task that comes before it (its immediate predecessor) and one task that comes after it (its immediate successor). A task begins when its predecessors have been completed. The start task has no predecessor, and the end task has no successor. These networks are called *connected*. They are the type of network used in this book. Figure 4-16 gives examples of how the variety of relationships that might exist between two or more tasks can be diagrammed.

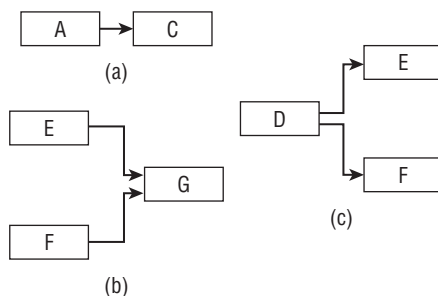


Figure 4-16 Diagramming conventions

Dependencies

A *dependency* is simply a relationship that exists between pairs of tasks. To say that task B depends on task A means that task A produces a deliverable that is needed in order to do the work associated with task B. There are four types of task dependencies, as illustrated in Figure 4-17.

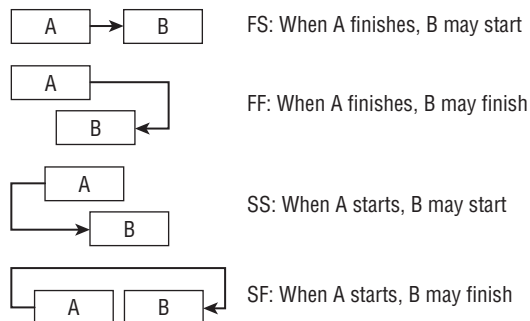


Figure 4-17 Dependency relationships

The four task dependencies shown in the figure are as follows:

Finish-to-start (FS) — This dependency says that task A must be complete before task B can begin. It is the simplest and most risk averse of the four types. For example, task A can represent the collection of data, and task B can represent entry of the data into the computer. To say that the dependency between A and B is finish-to-start means that once you have finished collecting the data, you may begin entering the data. I recommend using FS dependency in the initial project planning session. The FS dependency is displayed with an arrow emanating from the right edge of the predecessor task and leading to the left edge of the successor task.

Start-to-start (SS) — This dependency says that task B may begin once task A has begun. Note that there is a no-sooner-than relationship between task A and task B. Task B may begin no sooner than task A begins. In fact, they could both start at the same time. For example, you could alter the data collection and data entry dependency: As soon as you begin collecting data (task A), you may begin entering data (task B). In this case, there is an SS dependency between task A and B. The SS dependency is displayed with an arrow emanating from the left edge of the predecessor (A) and leading to the left edge of the successor (B). I use this dependency relationship in the “Compressing the Schedule” section later in the chapter.

Start-to-finish (SF) — This dependency is a little more complex than the FS and SS dependencies. Here task B cannot be finished sooner than task A has started. For example, suppose you have built a new information system. You don’t want to eliminate the legacy system until the new system is operable. When the new system starts to work (task A), the old system can be discontinued (task B). The SF dependency is displayed with an arrow emanating from the left edge of task A to the right edge of task B. SF dependencies can be used for just-in-time scheduling between two tasks, but they rarely occur in practice.

Finish-to-finish (FF) — This dependency states that task B cannot finish sooner than task A. For example, data entry (task B) cannot finish until data collection (task A) has finished. In this case, task A and B have a finish-to-finish dependency. The FF dependency is displayed with an arrow emanating from the right edge of task A to the right edge of task B. To preserve the connectedness property of the network diagram, the SS dependency on the front end of two tasks should have an accompanying FF dependency on the back end.

Constraints

The type of dependency that describes the relationship between tasks is determined as the result of *constraints* that exist between those tasks. Each type of constraint can generate any one of the four dependency relationships. The following four types of constraints will affect the sequencing of project tasks and, hence, the dependency relations between tasks:

- Technical constraints
- Management constraints
- Interproject constraints
- Date constraints

The following sections describe each of these constraint types in more detail.

Technical Constraints

Technical dependencies between tasks arise because one task (the successor) requires output from another (the predecessor) before work can begin on it. In the simplest case, the predecessor must be completed before the successor can begin. I advise using FS relationships in the initial construction of the network diagram because they are the least complex and risk-prone dependencies. If the project can be completed by the requested date using only FS dependencies, there is no need to complicate the plan by introducing other, more complex and risk-prone dependency relationships. SS and FF dependencies will be used later when you analyze the network diagram for schedule improvements.

Within the category of technical constraints, the following four related situations should be accounted for:

Discretionary constraints — These are judgment calls by the project manager that result in the introduction of dependencies. These judgment calls may be merely a hunch or a risk-aversion strategy taken by the project manager. Through the sequencing tasks, the project manager gains a modicum of comfort with the project work. For example, revisit the

data collection and data entry example used earlier in the chapter. The project manager knows that a team of recent hires will be collecting the data and that the usual practice is to have them enter the data as they collect it (SS dependency). The project manager knows that this introduces some risk to the process; and because new hires will be doing the data collection and data entry, the project manager decides to use an FS, rather than SS, dependency between data collection and data entry.

Best-practices constraints — These are based on past experiences that have worked well for the project manager or are known to the project manager based on the experiences of others in similar situations. The practices in place in an industry can be powerful influences here, especially in dealing with bleeding-edge technologies. In some cases, the dependencies that result from best-practices constraints, which are added by the project manager, might be part of a risk-aversion strategy following the experiences of others. For example, consider the dependency between software design and software build tasks. The safe approach has always been to complete the design before beginning the build. The current business environment, however, is one in which getting to the market faster has become the strategy for survival.

In an effort to get to market faster, many companies have introduced concurrency into the design-build scenario by changing the FS dependency between design and build to an SS dependency as follows. At some point in the design phase, enough is known about the final configuration of the software to begin limited programming work. By introducing this concurrency between designing and building, the project manager can reduce the time to market for the new software. Even though the project manager knows that this SS dependency introduces risk (design changes made after programming has started may render the programming useless), he or she will adopt this best-practices approach.

Logical constraints — These are like discretionary constraints that arise from the project manager's way of thinking about the logical way to sequence a pair of tasks. It's important for the project manager to be comfortable with the sequencing of work. After all, the project manager has to manage it. Based on past practices and common sense, you may prefer to sequence tasks in a certain way. That's acceptable, but do not use this as an excuse to manufacture a sequence out of convenience. As long as there is a good, logical reason, that is sufficient justification. For example, in the design-build scenario, several aspects of the software design certainly lend themselves to some concurrency with the build task. However, part of the software design work involves the use of a recently introduced technology with which the company has no experience. For that reason, the project manager

decides that the part of the design that involves this new technology must be complete before any of the associated build tasks can start.

Unique requirements — These constraints occur in situations where a critical resource — such as an irreplaceable expert or a one-of-a-kind piece of equipment — is involved on several project tasks. For example, suppose a new piece of test equipment will be used on a software development project. There is only one piece of this equipment, and it can be used on only one part of the software at a time. It will be used to test several different parts of the software. To ensure that no scheduling conflicts arise with the new equipment, the project manager creates FS dependencies between every part of the software that will use this test equipment. Apart from any technical constraints, the project manager may impose such dependencies to ensure that no scheduling conflicts will arise from the use of scarce resources.

Management Constraints

A second type of dependency arises as the result of a management-imposed constraint. For example, suppose the product manager on a software development project is aware that a competitor will soon introduce a new product with similar features. Rather than follow the concurrent design-build strategy, the product manager wants to ensure that the design of the new software will yield a product that can compete with the competitor's new product. He or she expects design changes in response to the competitor's new product and, rather than risk wasting the programmers' time, imposes the FS dependency between the design and build tasks.

You'll see management constraints at work when you analyze the network diagram and as part of the scheduling decisions you make as project manager. Dependencies based on management constraints differ from technical dependencies in that they can be reversed, whereas technical dependencies cannot. For example, suppose the product manager finds out that the competitor has discovered a fatal flaw as a result of beta testing and has decided to indefinitely delay the new product introduction pending resolution of the flaw. The decision to follow the FS dependency between design and build can now be reversed, and the concurrent design-build strategy can be reinstituted. That is, management will have the project manager change the design-build dependency from FS to SS.

Interproject Constraints

Interproject constraints result when deliverables from one project are needed by another project. Such constraints result in dependencies between the tasks

that produce the deliverable in one project and the tasks in the other project that require the use of those deliverables. For example, suppose a new piece of test equipment is being manufactured by the same company that is developing the software that will use the test equipment. In this case, the start of the testing tasks in the software development project depends on the delivery of the manufactured test equipment from the other project. The dependencies that result are technical but exist between tasks in two or more projects, rather than within a single project.

Interproject constraints arise when a very large project is decomposed into smaller, more manageable projects. For example, the construction of the Boeing 777 took place in a variety of geographically dispersed manufacturing facilities. Each manufacturing facility defined a project to produce its part. To assemble the final aircraft, the delivery of the parts from separate projects had to be coordinated with the final assembly project plan. Thus, there were tasks in the final assembly project that depended on deliverables from other sub-assembly projects.

NOTE These interproject constraints are common. Occasionally, large projects are decomposed into smaller projects or divided into a number of projects that are defined along organizational or geographic boundaries. In all of these examples, projects are decomposed into smaller projects that are related to one another. This approach creates interproject constraints. Although I prefer to avoid such decomposition because it creates additional risk, it may be necessary at times.

Date Constraints

At the outset, I want to make it clear that I do not approve of using date constraints. I avoid them in any way I can. In other words, “just say no” to typing dates into your project management software. If you have been in the habit of using date constraints, read on.

Date constraints impose start or finish dates on a task, forcing it to occur according to a particular schedule. In this date-driven world, it is tempting to use the requested date as the required delivery date. These constraints generally conflict with the schedule that is calculated and driven by the dependency relationships between tasks. In other words, date constraints create unnecessary complication in interpreting the project schedule.

The three types of date constraints are as follows:

No earlier than — Specifies the earliest date on which a task can be completed.

No later than — Specifies a date by which a task must be completed.

On this date — Specifies the exact date on which a task must be completed.

All of these date constraints can be used on the start or finish side of a task. The most troublesome application is the on-this-date constraint. It firmly sets a date and affects all tasks that follow it. The result is the creation of a needless complication in the project schedule and in reporting the status of the project as it progresses. The next most troublesome date constraint is the no-later-than constraint. It will not allow a task to occur beyond the specified date. Again, you are introducing complexity for no good reason. Both on-this-date and no-later-than constraints can result in negative slack. If at all possible, do not use them. There are alternatives, which are discussed in the next chapter.

The least troublesome date constraint is the no-earlier-than constraint. At worst, it simply delays a task's schedule and by itself cannot cause negative float.

Using the Lag Variable

Pauses or delays between tasks are indicated in the network diagram through the use of *lag variables*. Lag variables are best described with an example. Suppose that data is being collected by mailing out a survey and is entered as the surveys are returned. Imposing an SS dependency between mailing out the surveys and entering the data would not be feasible unless you introduced some delay between mailing surveys and getting back the responses that could be entered. For the sake of the example, suppose that you wait 10 days from the date you mail the surveys until you schedule entering the data from the surveys. Ten days is the time you think it will take for the surveys to arrive at the recipient locations, for the recipients to answer the survey questions, and for the surveys to be returned to you by mail. In this case, you have defined an SS dependency with a lag of 10 days. To put it another way, task B (data entry) can start 10 days after task A (mailing the survey) has started.

Creating an Initial Project Network Schedule

As mentioned, all tasks in the network diagram have at least one predecessor task and one successor task, with the exception of the start and end tasks. If this convention is followed, the sequence is relatively straightforward to identify. However, if the convention is not followed, if date constraints are imposed on some tasks, or if the resources follow different calendars, understanding the sequence of tasks that result from this initial scheduling exercise can be rather complex.

To establish the project schedule, you need to compute two schedules: the early schedule, which you calculate using the forward pass, and the late schedule, which you calculate using the backward pass.

The early schedule consists of the earliest times at which a task can start and finish. These are calculated numbers derived from the dependencies between

all the tasks in the project. The late schedule consists of the latest times at which a task can start and finish without delaying the completion date of the project. These are also calculated numbers that are derived from the dependencies between all of the tasks in the project.

The combination of these two schedules gives you the following two additional pieces of information about the project schedule:

- The window of time within which each task must be started and finished in order for the project to be completed on schedule
- The sequence of tasks that determine the project completion date

The sequence of tasks that determine the project completion date is called the *critical path*. The critical path can be defined in the following ways:

- The longest duration path in the network diagram
- The sequence of tasks whose early schedule and late schedule are the same
- The sequence of tasks with zero slack or float (as defined later in this chapter)

All of these definitions say the same thing: The critical path is the sequence of tasks that must be completed on schedule in order for the project to be completed on schedule.

The tasks that define the critical path are called *critical path tasks*. Any delay in a critical path task will delay the completion of the project by the amount of delay in that task. Critical path tasks represent sequences of tasks that warrant the project manager's special attention.

The *earliest start (ES)* time for a task is the earliest time at which all of its predecessor tasks have been completed and the subject task can begin. The ES time of a task with no predecessor tasks is arbitrarily set to 1, the first day on which the project is open for work. The ES time of tasks with one predecessor task is determined from the *earliest finish (EF)* time of the predecessor task. The ES time of tasks with two or more predecessor tasks is determined from the latest of the EF times of the predecessor tasks. The earliest finish (EF) time of a task is calculated as $((ES + \text{Duration}) - \text{One Time Unit})$. The reason for subtracting the one time unit is to account for the fact that a task starts at the beginning of a time unit (hour, day, and so forth) and finishes at the end of a time unit. In other words, a one-day task, starting at the beginning of a day, begins and ends on the same day. For example, in Figure 4-17 note that task E has only one predecessor: task C. The EF for task C is the end of day 3. Because it is the only predecessor of task E, the ES of task E is the

beginning of day 4. Conversely, task D has two predecessors: task B and task C. When there are two or more predecessors, the ES of the successor (task D in Figure 4-17) is calculated based on the maximum of the EF dates of the predecessor tasks. The EF dates of the predecessors are the end of day 4 and the end of day 3. The maximum of these is 4; therefore, the ES of task D is the morning of day 5. The complete calculations of the early schedule are shown in Figure 4-18.

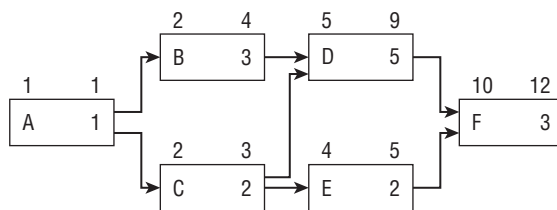


Figure 4-18 Forward-pass calculations

The *latest start (LS)* and *latest finish (LF)* times of a task are the latest times at which the task can start or finish without causing a delay in the completion of the project. Knowing these times is valuable for the project manager, who must make decisions regarding resource scheduling that can affect completion dates. The window of time between the ES and LF of a task is the window within which the resource for the work must be scheduled or the project completion date will be delayed. To calculate these times, you work backward in the network diagram. First set the LF time of the last task on the network to its calculated EF time. Its LS is calculated as $((LF - Duration) + \text{One Time Unit})$. Again, you add the one time unit to adjust for the start and finish of a task within the same day. The LF time of all immediate predecessor tasks is determined by the minimum of the LS, minus one time unit, times all tasks for which it is the predecessor.

For example, calculate the late schedule for task E in Figure 4-19. Its only successor, task F, has an LS date of day 10. The LF date for its only predecessor, task E, will therefore be the end of day 9. In other words, task E must finish no later than the end of day 9 or it will delay the start of task F and hence delay the completion date of the project. Using the formula, the LS date for task E will be $9 - 2 + 1$, or the beginning of day 8. Conversely, consider task C. It has two successor tasks: task D and task E. The LS dates for these tasks are day 5 and day 7, respectively. The minimum of those dates, day 5, is used to calculate the LF of task C — namely, the end of day 4. The complete calculations for the late schedule are shown in Figure 4-19.

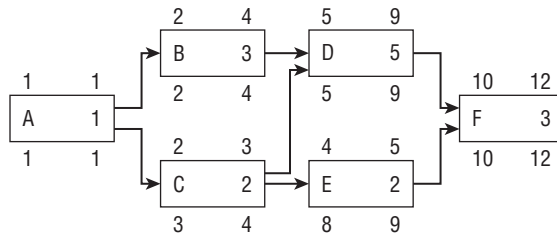


Figure 4-19 Backward-pass calculations

Critical Path

As mentioned, the critical path is the longest path or sequence of tasks (in terms of task duration) through the network diagram. The critical path drives the completion date of the project. Any delay in the completion of any one of the tasks in the sequence will delay the completion of the project. You should pay particular attention to critical path tasks. The critical path for the example problem you used to calculate the early schedule and the late schedule is shown in Figure 4-20.

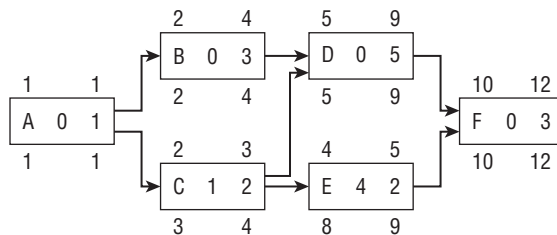


Figure 4-20 Critical path

Calculating Critical Path

One way to identify the critical path in the network diagram is to identify all possible paths through the diagram and add up the durations of the tasks that lie along those paths. The path with the longest duration time is the critical path. For projects of any size, this method is not feasible, and you have to resort to the second method of finding the critical path: computing the slack time of a task.

Computing Slack

This method of finding the critical path requires you to compute a quantity known as the task *slack time*. Slack time (also called *float*) is the amount of delay, expressed in units of time, that could be tolerated in the starting time or completion time of a task without causing a delay in the completion of the

project. Slack time is a calculated number. It is the difference between the late finish and the early finish ($LF - EF$). If the result is greater than zero, the task has a range of time in which it can start and finish without delaying the project completion date, as shown in Figure 4-21.

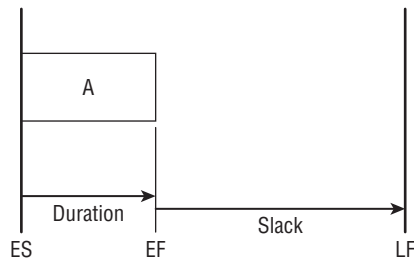


Figure 4-21 ES-to-LF window of a task

Because weekends, holidays, and other nonwork periods are not conventionally considered part of the slack, these must be subtracted from the period of slack.

There are two types of slack, as follows:

Free slack — This is the range of dates in which a task can finish without causing a delay in the early schedule of any tasks that are its immediate successors. Notice in Figure 4-20 that task C has an ES of the beginning of day 2 and an LF of the end of day 4. Its duration is two days, and it has a day 3 window within which it must be completed without affecting the ES of any of its successor tasks (task D and task E). Therefore, it has free slack of one day. Free slack can be equal to but never greater than total slack. When you choose to delay the start of a task, possibly for resource scheduling reasons, first consider tasks that have free slack associated with them. By definition, if a task's completion stays within the free slack range, it can never delay the early start date of any other task in the project.

Total slack — This is the range of dates in which a task can finish without delaying the project completion date. Consider task E in Figure 4-20. This task has a free slack (or float) of four days, as well as a total slack (or float) of four days. In other words, if task E were to be completed more than three days later than its EF date, it would delay completion of the project. If a task has zero slack, then it determines the project completion date. In other words, all the tasks on the critical path must be done on their earliest schedule or the project completion date will suffer. If a task with total slack greater than zero were to be delayed beyond its late finish date, it would become a critical path task and cause the completion date to be delayed.

Based on the method you used to compute the early and late schedules, the sequence of tasks having zero slack is defined as the critical path. If a task has been date-constrained using the on-this-date type of constraint, it will also have zero slack. However, this constraint usually gives a false indicator that a task is on the critical path. Finally, in the general case, the critical path is the path that has minimum slack.

Near-Critical Path

Even though project managers are tempted to rivet their attention on critical path tasks, other tasks also require their attention. These make up what is called a *near-critical path*. The full treatment of near-critical tasks is beyond the scope of this book. I introduce the concept here so that you are aware that paths other than critical paths are worthy of attention. As a general example, suppose the critical path tasks are tasks with which the project team has considerable experience. In this case, duration estimates are based on historical data and are quite accurate in that the estimated duration will be very close to the actual duration.

Conversely, suppose there is a sequence of tasks not on the critical path with which the team has little experience, so the duration estimates have large estimation variances. Also suppose that such tasks lie on a path that has little total slack. It is very likely that this near-critical path may actually drive the project completion date even though the total path length is less than that of the critical path. This situation will happen if larger-than-estimated durations occur. Because of the large duration variances, such a case is very likely. Obviously, this path cannot be ignored.

Analyzing the Initial Project Network Diagram

After you have created the initial project network diagram, one of the following two situations will be present:

- The initial project completion date meets the requested completion date. Usually this is not the case, but it does sometimes happen.
- The more likely situation is that the initial project completion date is later than the requested completion date. In other words, you have to find a way to squeeze some time out of the project schedule.

You eventually need to address two considerations: the project completion date and resource availability under the revised project schedule. The following section is based on the assumption that resources will be available to meet this compressed schedule. Later in this chapter, the resource-scheduling problem

is addressed. The two are quite dependent on one another, but they must be treated separately.

Compressing the Schedule

Almost without exception, the initial project calculations will result in a project completion date beyond the required completion date. That means the project team must find ways to reduce the total duration of the project to meet the required date.

To address this problem, you analyze the network diagram to identify areas where you can compress project duration. Look for pairs of tasks that allow you to convert tasks that are currently worked on in series into patterns of work that are more parallel. Work on the successor task might begin once the predecessor task has reached a certain stage of completion. In many cases, some of the deliverables from the predecessor can be made available to the successor so that work might begin on it.

WARNING Changing the predecessor after work has started on the successor creates a potential rework situation and increases project risk. Schedule compressions affect only the time frame in which work will be done; they do not reduce the amount of work to be done. The result is the need for more coordination and communication, especially between the tasks affected by the dependency changes.

You first need to identify strategies for locating potential dependency changes. Focus your attention on critical path tasks because these are the tasks that determine the completion date of the project, the very thing you want to have an impact on. You might be tempted to look at critical path tasks that come early in the life of the project, thinking that you can get a jump on the scheduling problem, but this usually is not a good strategy for the following reason: At the early stages of a project, the project team is little more than a group of people who have not worked together before (I refer to them as a herd of cats). Because you are going to make dependency changes (FS to SS), you are going to introduce risk into the project. Your herd of cats is not ready to assume risk early in the project. You should give them some time to become a real team before intentionally increasing the risk they will have to contend with. That means you should look downstream on the critical path for those compression opportunities.

A second factor to consider is focusing on tasks that are partitionable. A *partitionable* task is one whose work can be assigned to two or more individuals working in parallel. For example, painting a room is partitionable. One person

can be assigned to each wall. When one wall is finished, a successor task, such as picture hanging, can be done on the completed wall. That way, you don't have to wait until the room is entirely painted before you can begin decorating the walls with pictures.

Writing a computer program may or may not be partitionable. If it is partitionable, you could begin a successor task like testing the completed parts before the entire program is complete. Whether a program is partitionable depends on many factors, such as how the program is designed, whether the program is single-function or multifunction, and other considerations. If a task is partitionable, it is a candidate for consideration. You might be able to partition it so that when some of it is finished, you can begin working on successor tasks that depend on the part that is complete. After you have identified a candidate set of partitionable tasks, you need to assess the extent to which the schedule might be compressed by starting the task's successor task earlier. There is not much to gain by considering tasks with short duration times. I hope I have given you enough hints at a strategy to help you find opportunities to compress the schedule that you will be able to find those opportunities quite easily. If you can't, don't worry. The next chapter gives you other suggestions for compressing the schedule.

Assume you have found one or more candidate tasks to work with. See what happens to the network diagram and the critical path as dependencies are adjusted. As you begin to replace series (SF dependencies) with parallel sequences of tasks (SS dependencies), the critical path may change to a new sequence of tasks. This change will happen if, because of your compression decisions, the length of the initial critical path is reduced to a duration that is less than that of some other path. The result is a new critical path. Figure 4-22 shows two iterations of the analysis. The top diagram is the original critical path that results from constructing the initial network diagram using only FS dependencies. The critical path tasks are identified with a filled dot.

The middle diagram in Figure 4-22 is the result of changing the dependency between tasks A and B from FS to SS. Now the critical path has changed to a new sequence of tasks. The tasks with filled triangles illustrate the new critical path. If you change the FS dependency between tasks C and D, the critical path again moves to the sequence of tasks identified by the filled squares.

Occasionally, some tasks always remain on the critical path. For example, notice the set of tasks that have a filled circle, triangle, and square in Figure 4-22. They have remained on the critical path through both changes. This set of tasks identifies a bottleneck in the project schedule. Although further compression may result in this set of tasks changing, this set of bottleneck tasks does identify a set of tasks deserving of particular attention as the project commences. Because all critical paths generated to this point pass through this bottleneck, you might want to take steps to ensure that these tasks do not fall behind schedule.

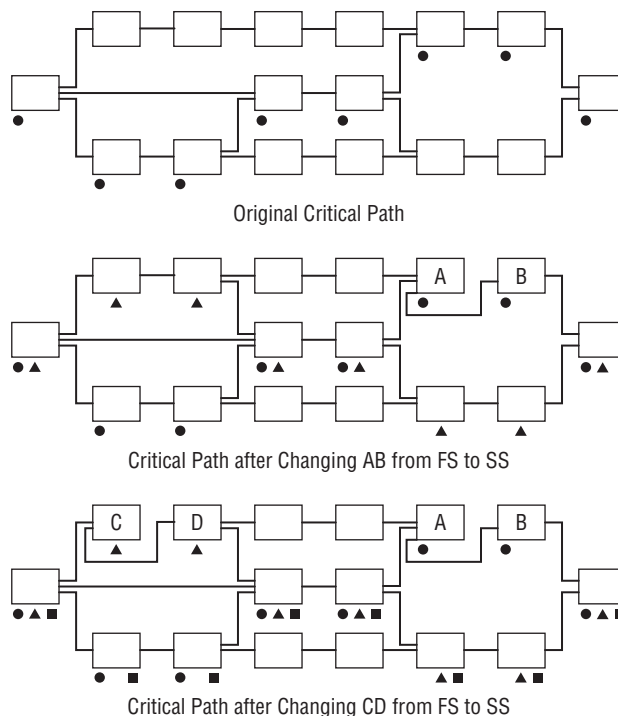


Figure 4-22 Schedule compression iterations

Management Reserve

Management reserve is a topic associated with task duration estimates, but it more appropriately belongs in this section because it should be a property of the project network more so than of the individual tasks.

At the individual task level, you might be tempted to pad your estimates to have a better chance of finishing a task on schedule. For example, you know that a particular task will require three days of your time to complete, but you submit an estimate of four days just to make sure you can get the three days of work done in the four-day schedule you hope to get for the task. The one day that you add is padding. First, let's agree that you will not do this. Parkinson's Law (which states that work will expand to the time allotted to complete it) will surely strike you down, and the task will, in fact, require the four days you estimated it would take. Stick with the three-day estimate and work to make it happen. That is a better strategy. Now that you know padding is bad at the task level, you are going to apparently contradict yourself by saying that it is all right at the project level. There are some very good reasons for this.

Management reserve is nothing more than a contingency budget of time. The size of that contingency budget can be in the range of 5 to 10 percent of the total of all the task durations in your project. The size might be closer to 5 percent for projects having few unknowns, or it could be closer to 10 percent for projects using breakthrough technologies or that are otherwise very complex. After you have determined the size of your management reserve, you create a task whose duration is the size of management reserve and put that task at the end of the project. It will be the last task, and its completion will signal the end of the project. This management reserve task becomes the last one in your project plan, succeeded only by the project completion milestone.

What is this management reserve used for? First, the project team should manage the project so that the reserve task is not needed — though in reality, this is rarely possible. The date promised to the client is the one calculated by the completion of the reserve task. The reserve task's duration can be shortened as necessary. For example, if the critical path slips by two days, the reserve task's duration will be reduced by two days. This holds the project completion date constant.

This technique keeps the management reserve task visible and enables you to manage the rate at which it's being used. If 35 percent of the overall project time line has elapsed and 50 percent of the reserve task has been used, you know you're heading for trouble.

Second, management reserve can be used as incentive for the project team. For example, many contracts include penalties for completing milestones later than planned, as well as rewards for completing milestones ahead of schedule. Think of management reserve as a contingency fund that you do not want to spend. Every day that is left in the contingency fund at the completion of the project is a day ahead of schedule for which the client should reward you. Conversely, if you spend that contingency fund and still require more time to complete the project, this means that the project was completed later than planned. For every day that the project is late, you should expect to pay a penalty.

Planning for Project Risk: The Risk Management Life Cycle

Every project is subject to risks. Some can be identified and plans can be put in place if they occur; others cannot and must be dealt with as they occur. The events that this section focuses on are those that could compromise the successful completion of the project. No one knows when they will occur, but they will occur with some likelihood and cause some damage to the project. For example, the loss of a team member who has a critical or scarce skill is one

such event. The longer the project lasts, the more likely this will happen. The history of some organizations might suggest that this is a certainty. Knowing this, what would you do? That is the question answered this section. The answer lies in understanding what the risk management life cycle is and how to construct a risk management plan.

The risk management life cycle was introduced and briefly explained in Chapter 2. It has the following four phases:

- Risk identification
- Risk assessment
- Risk mitigation
- Risk monitoring and control

These are discussed in detail in the sections that follow.

Risk Identification

In order to establish a risk management program for the project, there are several processes through which the project manager and project team must go. The first is risk identification, and it generally occurs as part of project planning activities. In this part of the process, the entire planning team is brought together to discuss and identify the risks that are specific to the current project.

Developing a risk management plan is a significant part of the project planning process. The more complex and uncertain the project, the more important it is to have a dynamic and maintained risk management plan. Some have said that the project manager does nothing more than manage risk on the project. That is too restrictive, but it does speak to the importance of a good risk management plan for every project. Although the experienced project manager will certainly know what general type of risks there are on each project, the professional project manager takes nothing for granted and always engages the project planning team in identifying risks for the project. The list of risks can be cumulatively developed in parallel with other project planning activities. After that list is built, the team can move to the second step in the risk management process.

Risk Categories

There are four risk categories. Each category is listed here with its potential risks. Use these as suggestions only. Your specific project will probably suggest these or other risks within each of these categories. The planning team should review each list and modify it as needed.

Technical Risks

These may include the following:

- Quality and performance goals generally relating to the technology of the project
- The suitability, reliability, and quality or performance standards surrounding the technology
- Technology availability and complexity issues

Project Management Risks

These may include the following:

- Poor allocation of the project's resources
- Inadequate project management structure — proper planning processes to define critical deliverables for each project phase
- Inadequate planning, resource inexperience, or poor use of management disciplines
- Cost and schedule risks due to the aforementioned project management risks

Organizational Risks

These may include the following:

- Supportability risks or inadequate prioritization of projects
- Inadequacy of or interrupted funding and/or resource assignments
- Conflicts with other competing projects
- Policies that do not support efficient management and could potentially introduce supportability risks
- Politics and agendas that impede the development of the project's executing objectives

External Risks

These may include the following:

- Shifting legal or regulatory requirements
- Supplier and contractor risks and/or contract issues
- Economic collapse or work stoppages (strikes)
- Programmatic or supportability risks caused by external parties
- Deliverables from teams that are external to your own (IT or client)

Risk Assessment Template

Figure 4-23 shows a template that you can use for defining risks in each of these categories and making a preliminary assessment of how they might impact the scope matrix.

RISK CATEGORIES AND RISKS	SCOPE TRIANGLE ELEMENTS				
	Scope	Time	Cost	Quality	Resources
Technical					
Project Management					
Organizational					
External					

Figure 4-23 Risk identification and assessment template

Candidate Risk Drivers

The first step in the Risk Management Process is to identify the risk drivers that may be operative on a given project. These are the conditions or situations that may unfavorably affect project success. As an example, Figure 4-24 shows a candidate list from which the list of risk drivers that are appropriate for a given project can be chosen.

Risk Assessment

As mentioned previously, there are two major factors in assessing risk. The first of these is the probability that the risk event will occur. For instance, if a project involves migrating legacy systems to new systems, the interface points between the two are often where problems occur. The professional project manager will have a good sense of these types of risks and how much of a chance there is that they will occur. (By the way, if you are certain that an event will occur, it's not a risk, it's a certainty. This type of event isn't handled by risk management, because you are sure that it will occur. This means no probability is involved. No probability, no risk.)

Prioritize (from 1 to 10) the top 10 risk drivers for your project.

☐ Schedule is too aggressive	☐ Inadequately skilled personnel
☐ Overambitious performance	☐ Continuous requirements changes
☐ Too conservative a budget	☐ Inadequate development plan
☐ Unrealistic expectations	☐ Unsuitable organizational structure
☐ Misunderstood contract terms	☐ Testing facilities not available
☐ New/unfamiliar technology	☐ Poor software engineering methods
☐ Inadequate software sizing	☐ Poor technology support
☐ Unsuitable development model	☐ Lack of political support for project
☐ Unfamiliar new hardware	☐ Inconsistent client involvement
☐ Poorly defined requirements	☐ Loss of critical team member
☐ Frequent change requests	☐ Vendor/contractor relations
☐ Poorly defined processes	☐ Market/competitor pressures
☐ Volatile business environment	

Figure 4-24 Candidate risk drivers

When the team puts together the risk identification list, nothing should be ruled out at first. Let the team brainstorm risk without being judgmental. Some risks are so small that you will eventually ignore them. For instance, the risk that a meteor will destroy the building in which you work is miniscule. If you're worrying about things like this, you won't be much of a project manager. You need to manage the risks that actually might occur.

The second part of risk assessment is the expected loss the risk will have on the project. If the probability is high and the impact is low, you may be able to ignore the risk. If the probability is low but the impact is high, you might also be able to ignore the risk. The decision is based on the product of the probability of the event happening and the impact it will have. For example, if the probability of losing a critical skill is 0.8 (probability is a number between 0 and 1.0) and the impact is \$50,000, the expected loss is \$40,000 ($0.8 \times \$50,000$). As a further example, suppose the probability of the Bull on Wall Street being stolen is 1×10^{-10} and the impact is \$75,000,000; then the expected loss is \$750.

You should ignore the risk if the cost of avoiding the risk is greater than the expected loss. In other words, don't solve a \$100 problem with a \$1,000 solution. In the two examples, you would most likely not ignore the risk of losing the critical skill, but you would ignore the risk of the Bull on Wall Street being stolen.

Static Risk Assessment

If you don't want to get hung up on numeric risk assessments, you might want to try using the risk matrix shown in Figure 4-25. There is nothing magic about using a 3×3 matrix. A 5×5 matrix works just as well.

		Probability		
		L	M	H
Loss	L	IGNORE	IGNORE	CONSIDER
	M	IGNORE	CONSIDER	TAKE ACTION
	H	CONSIDER	TAKE ACTION	TAKE ACTION

Figure 4-25 Risk matrix

For each risk, evaluate the probability that it will occur on a Low, Medium, High scale and the impact on a Low, Medium, High scale. The combination of these two assessments identifies a specific cell in the risk matrix with the recommended action if any. The situation regarding this risk may change later in the project. So, my advice is to monitor the risk, but don't act unless reason dictates that you do so.

Dynamic Risk Assessment

The preceding risk assessment is basically static. By that I mean an analysis is done during planning, and a risk management plan is put in place for the entire project. It does not change as the project progresses. That is the simplest approach and probably less effective than the dynamic risk assessment discussed in this section. I have used the following dynamic risk assessment approach with great success. In this approach, risk is continuously reassessed at each phase of the project. An example will help explain how this approach is used.

After the risk drivers have been identified, they must be ranked from most likely to have an impact on the project to least likely to have an impact on the project. Label them A (most likely) through J (least likely) and array the data as shown in Figure 4-26. The column entries are 1 = low risk, 2 = medium risk, and 3 = high risk. Actually, any metric can be used as long as the lower numbers are at the low-risk end and the higher numbers are at the high-risk end.

The data given in the worksheet is from a hypothetical project. The columns are the top risk drivers that were identified from the candidate list, and the rows are steps in a process. For the sake of an example, I chose steps from a

hypothetical systems-development life cycle. Any collection of process steps may be used, so the tool has broad application for a variety of contexts. A score of 1 is given to risk drivers that will not impact the process step if they should occur, 2 is for a medium impact, and 3 is for a strong impact. Actually, any numeric scale may be used. The row and column totals are evaluated relative to one another and to scores from similar projects. These totals tell the story. High column totals suggest a risk driver that is operative across a number of steps in the process. High row totals suggest a process step that is affected by several risk drivers. Finally, the total for the whole worksheet gives you a percentage that can be used to compare this project against similar completed projects. The percentage is relative, but it may suggest a rule that provides an early warning of projects that are high risk overall.

Project Activity	A	B	C	D	E	F	G	H	I	J	Score
Rqmnts Analysis	2	3	3	2	3	3	2	2	1	1	22
Specifications	2	1	3	2	2	2	1	2	2	3	20
Preliminary Design	1	1	2	2	2	2	1	2	2	2	17
Design	2	1	2	2	2	3	1	2	2	1	18
Implement	1	2	2	3	3	2	1	2	2	1	19
Test	2	2	2	2	2	3	2	2	2	2	21
Integration	3	2	3	3	3	3	2	3	3	2	27
Checkout	1	2	2	3	3	3	2	3	2	2	23
Operation	2	2	3	3	3	3	3	3	1	1	24
Score	16	16	22	22	23	24	15	21	17	15	191

Maximum score is 270. Risk level for this project is $191/270 = 71\%$.

Figure 4-26 Risk assessment worksheet

To analyze the resulting scores, first examine column totals that are large relative to other column totals. In the example, you should focus on the risk drivers associated with columns C, D, E, and F. Because their column totals are high, they can potentially affect several process steps. The project team should identify strategies for either reducing the probability of the risk occurring or mitigating its impact, or both, should the event associated with that risk occur. The row totals can be analyzed in the same fashion. In the example, integration has the highest row total (27). This indicates that several risk drivers can impact integration. The project team should pay attention to the work associated with integration and look for ways to improve or better manage it. For example, the team might choose to have more skilled personnel work on integration than they might otherwise choose.

In the example, the risk factor is 70 percent. This value can be interpreted only in comparison to the risk factor of completed projects. There will be a

pattern of project failures for projects whose risk factor is above a certain number. If 70 percent is above that number, the example project is a high risk for failure. The decision to do this project will have to be offset by the business value the project expects to contribute.

Risk Mitigation

The next step in risk management is to plan, as much as possible, the responses that will be used in the event that the identified risks occur. For instance, you may want to include a clause in your hardware contract with the vendor that if you don't receive the servers by a certain date, then the vendor will pay a penalty. This penalty gives the vendor an incentive to analyze and mitigate the risks involved in late delivery of key equipment. For all the risks listed in the risk identification that you choose to act upon, you should have some type of mitigation strategy in mind. It's not enough simply to list the risks; you need to plan to do something about the risk events should they occur.

Another example of risk planning is planning for key personnel. What will you do if one of the key developers leaves the company before finishing the coding? This risk will impact the project severely if it occurs. Having someone capture code as it is written and debriefing with the developer each day are two ways of dealing with the risk of key personnel loss. How many others can you come up with? Coming up with contingency plans such as these is good risk response planning.

In order to build an effective risk mitigation plan, you should consider the following five risk responses:

- **Accept** — Do nothing because the cost of the action is more expensive than the expected loss.
- **Avoid** — Elect not to do the part of the project associated with the risk.
- **Contingency planning** — Frame plans to deal with the risk consequence and monitor risk regularly (identify contingency trigger points).
- **Mitigate** — Reduce the probability of occurrence, the impact, or both.
- **Transfer** — Outsource.

Risk Monitoring and Control

After you've identified the risk, assessed the probability and impact of the risks, and planned what you will do if the risk even occurs, then you need to monitor and control the project risks. Writing down the risks and assessing them gives everyone on the project team an awareness of their existence. This is a start. To accommodate the monitoring and control phase of the risk

management life cycle, you should maintain a risk log. This document lists all risks that you want to manage and describes what the risk is, who is supposed to manage the risk, and what has been done to manage the risk event.

A risk log is a simple template that you can create in Microsoft Word. A typical risk log will contain the following five fields:

- **ID number** — This always remains the same, even if the risk event has occurred and been managed. If you take the risk off the list and file it elsewhere, don't assign the old number to a new risk. Keep the original number with the discarded risk and never use it again, or there will be a great deal of confusion.
- **Risk description** — This is a short statement of the risk event.
- **Risk owner** — This is the person who has the responsibility of monitoring the status of the listed risk.
- **Action to be taken** — Lists what the risk owner is going to do to deal with the risk event.
- **Outcome** — Describes what happened as a result of your mitigation strategy

Use the risk log to keep track of risks in the project and maintain control over them. When you go to status meetings, you should always talk about risks and their management by the team. Keep the risks in front of the team so that each member will be aware of what risks are coming up and what is to be done about the risk event. This is just exercising common sense and therefore good project management.

Writing an Effective Project Proposal

The deliverable from all the planning activities in the JPPS is the project proposal. It is the document you will forward to the senior management team for approval to do the project. In most cases, this will be the same team that approved the project for planning based on the POS. The project proposal states the complete business case for the project. This includes the expected business value, as well as cost and time estimates. In addition to this information, the proposal details what is to be done, who is going to do it, when it is going to be done, and how it is going to be done. It is the road map for the project.

NOTE Expect feedback and several revisions before approval is granted. It is not the purpose of this section to spell out in detail what a project proposal should look like. The organization will have a prescribed format to follow. This section merely outlines the contents you will be expected to submit.

Contents of the Project Proposal

Each organization will have a prescribed format for its project proposal, but most proposals have sections similar to the ones listed in the sections that follow. The project proposal is a restatement of all the planning work that has been done so far.

Executive Summary

This section may not exceed one page and in most cases should be about a half page. Think of the two-minute elevator speech (if you can't summarize the project in a two-minute elevator ride, you haven't done your job) and you won't go wrong. I recommend that this section include three brief paragraphs, each describing one of the following topics:

- Business situation (expanded from the POS)
- Your project goal (expanded from the POS)
- Business value (expanded from the POS)

Now that was easy wasn't it?

If there is a strategic plan in place for your organization, you might want to add a fourth paragraph that briefly describes how your project supports that strategic plan. This will be necessary if your project is competing with other projects for a place in the project portfolio. See Chapter 14 for more details on this important topic.

Background

This is a brief description of the situation that led to the project proposal. It often states the business conditions, opportunities, and any problems giving rise to the project. It sets the stage for later sections and puts the project in the context of the business.

Objective

This is another short section that gives a very general statement of what you hope to accomplish through this project. Avoid jargon — you don't know who might have reason to read this section. Use the language of the business, not the technical language of your department. The objective should be clearly stated so that there is no doubt as to what is to be done and what constitutes attainment of the objective.

Overview of the Approach to Be Taken

For those who might not be interested in the details of how you are going to reach your objective, this section provides a high-level outline of your

approach. Some mention of the PMLC model to be used would be good here. Again, avoid jargon whenever possible. Give a brief statement of each step and include a few sentences of supporting narrative. Brevity and clarity are important.

Detailed Statement of the Work

Here is where you provide a high-level summary of what will be done, when it will be done, who will do it, how much time will be required, and what criteria will be used to measure completeness. This is the road map of all the project work. Gantt charts are useful for such presentations of schedule data because they are easily understood and generally intuitive, even for people who are seeing them for the first time.

Time and Cost Summary

It is my practice to include a summary page of time and cost data. This usually works best if presented as a single high-level table. Often the data will have been stated over several pages, and it is brought together here for easy review and comment by the client.

Appendices

I recommend reserving the appendix for all supporting data and details that are not germane to the body of the proposal. Anticipate questions your client might have, and include answers here. Remember that this is detail beyond the basic description of the project work. Supporting information is generally found here.

Format of the Project Proposal

There are no hard-and-fast rules regarding the format of a project proposal. You will surely be able to find examples of successful proposals in your department or company that you can use as guides. After you have your ideas sketched out, share the proposal with a trusted colleague. His or her feedback may be the most valuable advice you can get.

Gaining Approval to Launch the Project

Getting your POS approved means that the senior management team thinks your idea addresses an important business situation and that they will give you the resources you need to develop your project plan. Your project proposal

has to convince them that your approach makes good business sense and the resources you are requesting are in line with the business value that will be generated. To gain that approval, you may have to submit and revise the proposal several times. That will come about for one or more of the following reasons:

- **The cost-benefit is not in your favor.** Decompose your solution and estimate benefits by function. Perhaps you can trim the solution down by eliminating functions with unfavorable cost-benefit ratios.
- **The risks of failure are too high.** This happens often. For example, a new technology or one for which your organization is not experienced is the major contributor to high risk. Replace that technology with a more stable, well-known technology and wait for time to correct the situation.
- **The total project cost exceeds available funding.** The scope triangle would suggest trimming scope or decomposing the project into phases.
- **Other projects are competing for the same resources.** Maybe it's time to press your sponsor into service. The politics or the leverage your sponsor can bring to bear may be sufficient.

For any or all of these reasons, you may be asked to revise and resubmit your proposal.

Putting It All Together

The work of planning the project is now complete. You have written and submitted a detailed project proposal based on the project plan. To the best of your ability, you've provided a timeline, budget, and resource requirements list. Approval at this stage means approval to do the project as defined in the project plan. In the next chapter, armed with an approved project proposal, project plan, and the resources to support the project, you will begin recruiting team members and putting the necessary team operating rules in place.

Discussion Questions

1. What are the advantages and disadvantages of holding a JPPS session onsite versus offsite?
2. Your planning session seems to have reached an impasse. The planning team is divided between two ways to approach a particularly difficult part of the project. Approximately two-thirds of the team members

want to use a well-tested and well-understood approach. The remaining third (of which you are a member) wants to use a new approach that holds the promise of significantly reducing the time to complete this part of the project. You are the project manager and feel very strongly about using the new approach. Should you impose your authority as project manager and take the new approach, or should you go with the majority? What is the basis for your decision? Be specific. Is there anything else you might do to resolve the impasse?

3. The WBS identifies all of the work that must be done to complete the project. What would you do if the answer to a question posed as part of the work determines which of the two alternatives mentioned in the previous question you should pursue?
4. Under what conditions might you choose to decompose an activity that meets all of the six completeness criteria? Give specific examples.
5. Can you think of any activities that would not meet all six completeness criteria, yet need not be further decomposed? Give specific examples.
6. You have used the three-point method to estimate the duration of a task that you know will be critical to the project. The estimate produces a very large difference between the optimistic and pessimistic estimates. What actions might you take, if any, regarding this task?
7. Discuss a project on which you've worked where time was the major factor in determining the success or failure of the project. What did you do about cost considerations? Did the sponsor(s) agree with the added cost? Was the project successful?
8. Prepare a simple budget showing an order of magnitude estimate, a budget estimate, and a definitive estimate. What did you have to do to bring each successive budget closer to the final working budget?
9. The project network diagram has been constructed, and the project completion date is beyond the management-imposed deadline. You have compressed the schedule as much as possible by introducing parallel work through changes from FS to SS dependencies, and you still do not meet the schedule deadline. What would you do? (Hint: Use the scope triangle discussed in Chapter 1.)
10. Even though all of your tasks have met the WBS completion criteria, what scheduling problems might prompt you to further decompose one or more of them, and how will that resolve the problems?
11. You are the project manager of a project to develop a new system for the company. You have two options for a resource to work on a

specific programming task for your project. The task is not on the critical path, but it is somewhat complex. Your options are as follows:

- (a) One choice is Harry. He is the most skilled programmer in the company and is therefore in constant demand. As a result, he is usually assigned to several projects at the same time. He is available to your project on a half-time basis. He currently has commitments to two other projects for the remaining half of his time.
- (b) Your other choice is actually a team of two programmers, both of whom have average skills. They are recent hires into the company and have never worked together before. You have two alternatives and are free to choose whichever one you want. First, you could pick one of these programmers to work half-time on your project. Second, they could each be assigned to your project quarter-time. Regardless of which choice you make, this would be the only project they would be working on. The remainder of their time will be spent in training on company processes and systems, and orientation to company policies and practices.

You have three alternatives from which to choose. Identify and evaluate the advantages and disadvantages of each alternative. What are the risks associated with each alternative? How might you mitigate those risks? Which choice would you make and why? Are there conditions under which one choice would be preferred over the other? Be specific.

CASE STUDY – PIZZA DELIVERED QUICKLY (PDQ)

12. The PDQ system consists of the following six subsystems:

- **Pizza Factory Locator**
- **Order Entry**
- **Logistics**
- **Order Submit**
- **Routing**
- **Inventory Management**

Pick one of these subsystems and build a complete WBS. You may have to make assumptions in order to complete this exercise. If so, just state them with your rationale.